

Name: _____



New York State Testing Program

Mathematics Test Session 1

Grade 8

Spring 2026

RELEASED QUESTIONS

Grade 8 Mathematics Reference Sheet

CONVERSIONS

1 yard = 3 feet
1 mile = 5,280 feet

1 cup = 8 fluid ounces
1 pint = 2 cups
1 quart = 2 pints
1 gallon = 4 quarts

1 pound = 16 ounces
1 ton = 2,000 pounds

CONVERSIONS ACROSS MEASUREMENT SYSTEMS

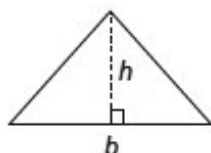
1 inch = 2.54 centimeters
1 meter = 39.37 inches
1 mile = 1.609 kilometers
1 kilometer = 0.6214 mile

1 gallon = 3.785 liters
1 liter = 0.2642 gallon

1 pound = 0.454 kilogram
1 kilogram = 2.2 pounds

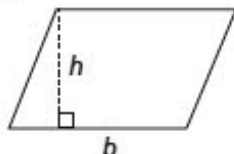
FORMULAS AND FIGURES

Triangle



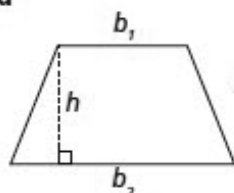
$$A = \frac{1}{2}bh$$

Parallelogram



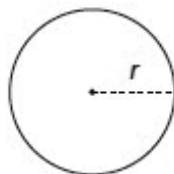
$$A = bh$$

Trapezoid



$$A = \frac{1}{2}h(b_1 + b_2)$$

Circle

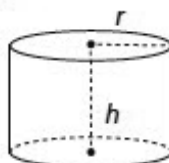


$$\begin{aligned}C &= 2\pi r \\C &= \pi d \\A &= \pi r^2\end{aligned}$$

General Prism

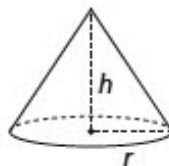
$$V = Bh$$

Right Cylinder



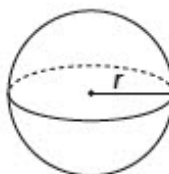
$$V = \pi r^2 h$$

Right Cone



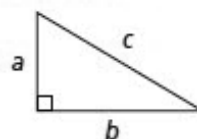
$$V = \frac{1}{3}\pi r^2 h$$

Sphere



$$V = \frac{4}{3}\pi r^3$$

Pythagorean Theorem



$$c^2 = a^2 + b^2$$

Session 1



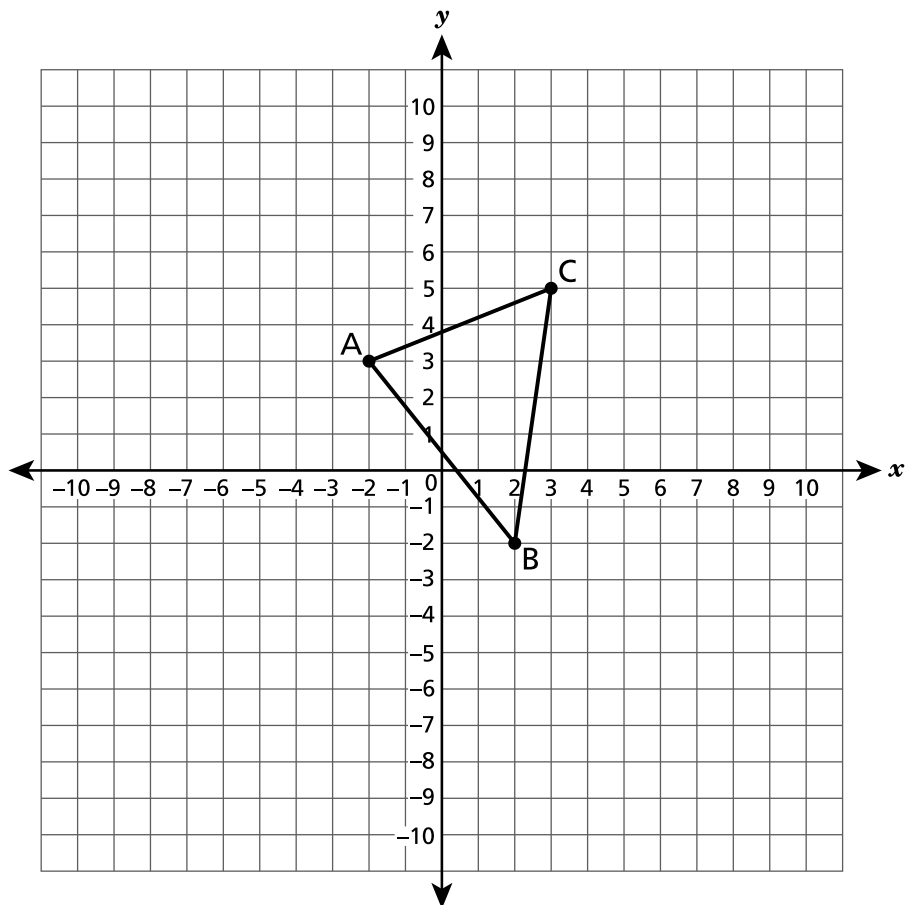
TIPS FOR TAKING THE TEST

Here are some ideas to help you do your best:

- Read each question carefully. Take your time.
- You have a ruler, a protractor, a reference sheet, and a calculator that you can use on the test if they help you answer the question.

1

Triangle ABC is graphed on the coordinate plane shown below. Triangle ABC will be translated 4 units to the left and 2 units up to create triangle $A'B'C'$.



What will be the coordinates of vertex A' ?

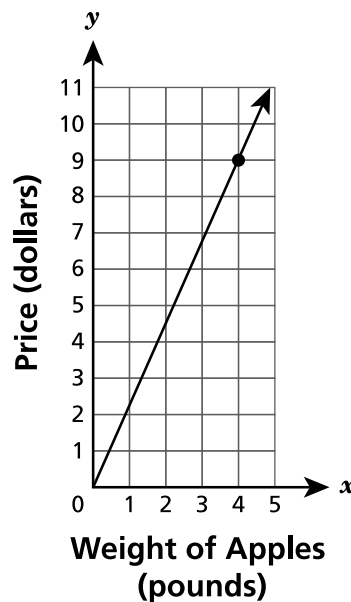
- A $(0, -1)$
- B $(2, 1)$
- C $(-1, 7)$
- D $(-6, 5)$

GO ON

Ms. Whitley is comparing the prices of apples, sold by the pound, at two different stores.

- The price, y , in dollars, of x pounds of apples sold at Store A is represented by the equation $y = 1.50x$.
- The price for a number of pounds of apples sold at Store B is represented by the graph shown below.

STORE B APPLE PRICES



Which statement is true based on the information given?

- A** Store A charges \$0.75 more per pound.
- B** Store B charges \$0.75 more per pound.
- C** Store A charges \$3.00 more per pound.
- D** Store B charges \$3.00 more per pound.

3

Which expression is equivalent to $\frac{12^{20}}{12^4}$?

A 1^5

B 1^{16}

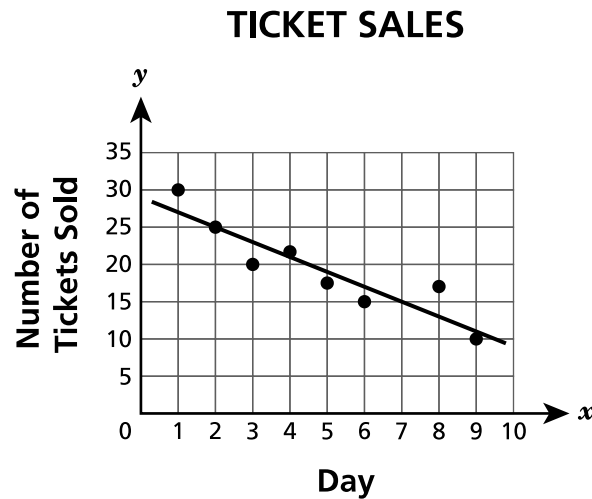
C 12^5

D 12^{16}

GO ON

4

A school sells tickets to a play. The data on the graph below shows the number of tickets sold, y , per day, x . The equation of the line $y = -2x + 29$ was drawn to best represent the relationship between the data.



Based on the equation of the line, which statement describes a valid prediction of ticket sales?

- A Since the number of tickets approximately decreases by 29 a day, it can be predicted that 9 tickets will be sold on day 12.
- B Since the number of tickets approximately decreases by 29 a day, it can be predicted that 5 tickets will be sold on day 12.
- C Since the number of tickets approximately decreases by 2 a day, it can be predicted that 9 tickets will be sold on day 12.
- D Since the number of tickets approximately decreases by 2 a day, it can be predicted that 5 tickets will be sold on day 12.

GO ON

5

Which value is irrational?

A 0

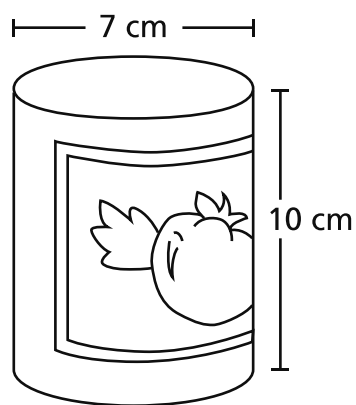
B π

C $\sqrt{9}$

D $\frac{1}{3}$

6

The dimensions of a can of tomatoes are shown in the diagram below.



What is the volume, to the nearest cubic centimeter, of the can?

A 110

B 220

C 385

D 1,539

GO ON

7

Triangle ABC has side lengths of 9 in., 12 in., and 15 in. Which equation proves that the triangle is a right triangle?

A $\sqrt{9} + \sqrt{12} = 15$

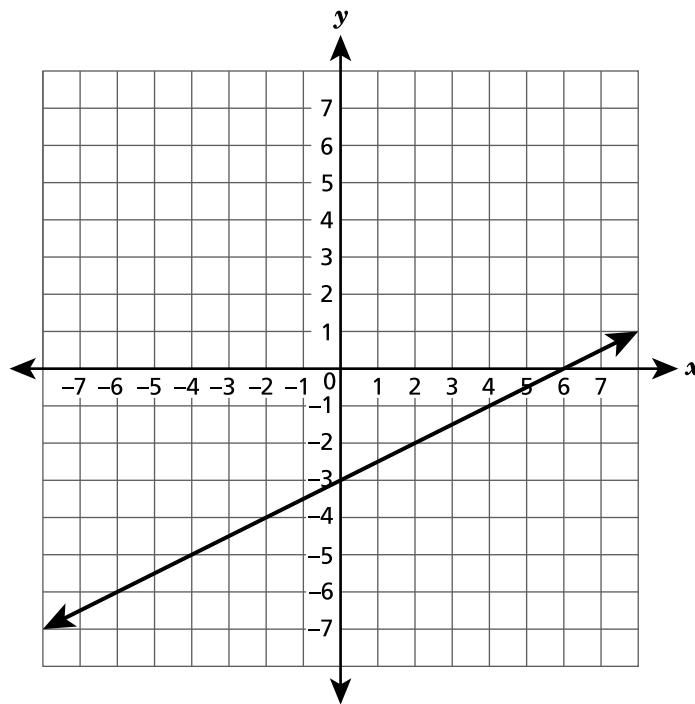
B $\sqrt{9} + \sqrt{12} = \sqrt{15}$

C $(9 + 12)^2 = 15^2$

D $9^2 + 12^2 = 15^2$

GO ON

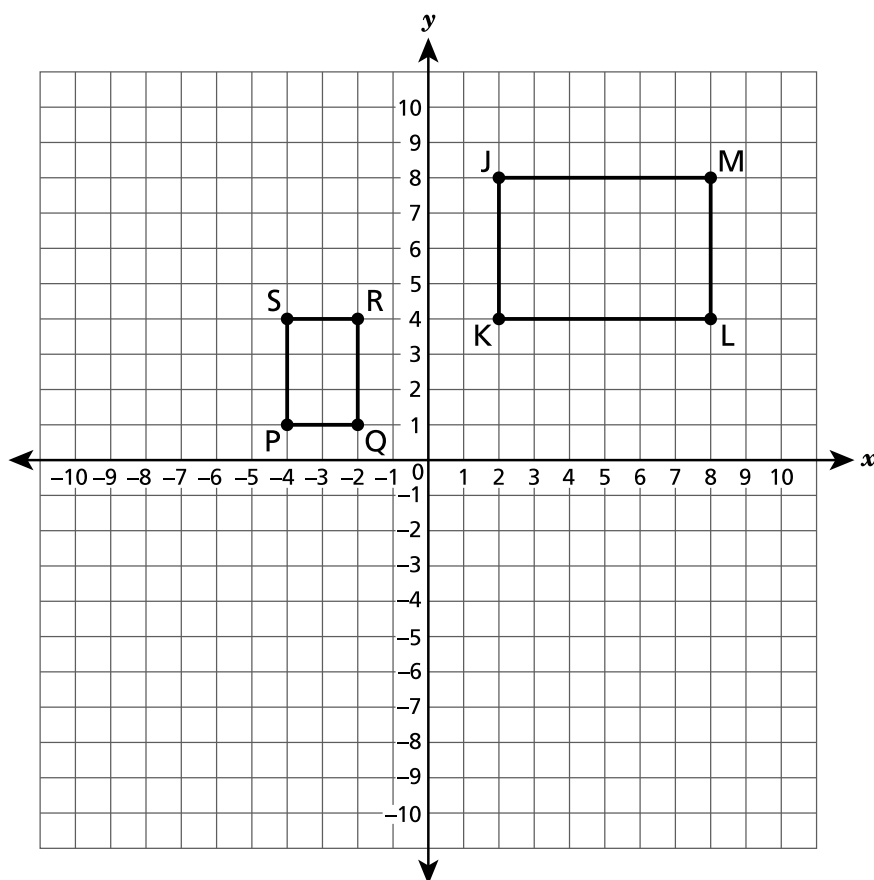
A line is graphed on the coordinate plane shown below.



Which equation represents the line shown on the coordinate plane?

- A $y = \frac{1}{2}x - 3$
- B $y = 3x - \frac{1}{2}$
- C $y = -3x + 6$
- D $y = -\frac{1}{2}x + 3$

Rectangles JKLM and PQRS are shown on the coordinate plane below.

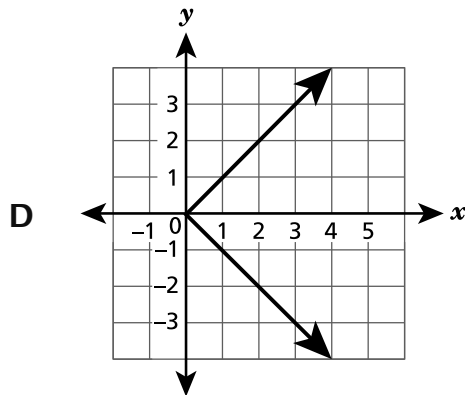
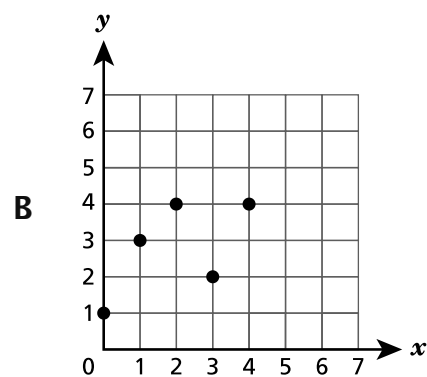
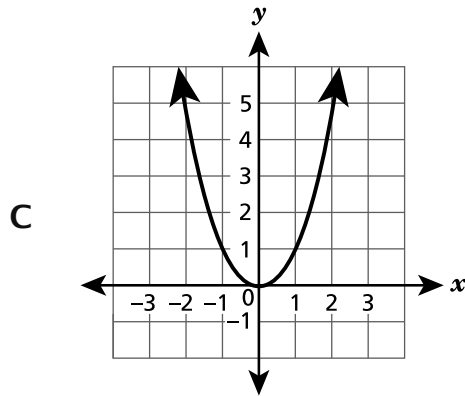
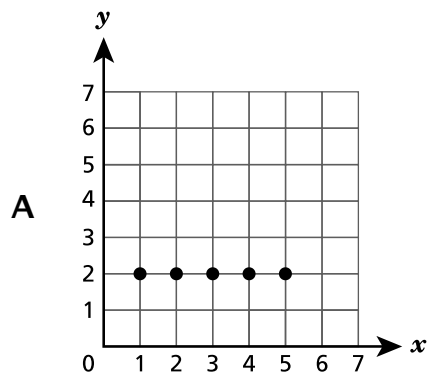


Which sequence of transformations can be used to show rectangle JKLM is similar but not congruent to rectangle PQRS ?

- A A reflection of rectangle JKLM over the y axis, followed by a dilation by a scale factor of 2 centered at the origin.
- B A reflection of rectangle JKLM over the y axis, followed by a dilation by a scale factor of $\frac{1}{2}$ centered at the origin.
- C A 90° counterclockwise rotation of rectangle JKLM about the origin, followed by a dilation by a scale factor of 2 centered at the origin.
- D A 90° counterclockwise rotation of rectangle JKLM about the origin, followed by a dilation by a scale factor of $\frac{1}{2}$ centered at the origin.

GO ON

11

Which relation is **not** a function?

12

Which equation has an infinite number of solutions?

A $2x - 1 = 2(x - 1)$

B $2x - 2 = 2(x - 1)$

C $2x - 3 = 3(x - 1)$

D $2x + 3 = 3(x - 1)$

GO ON

13

Which side lengths, in inches, will form a triangle?

A 3, 5, 10

B 4, 9, 14

C 5, 7, 10

D 6, 7, 14

14

Which equation represents a linear function?

A $y = 2x$

B $y = 2x^3$

C $y = x^2 - 2$

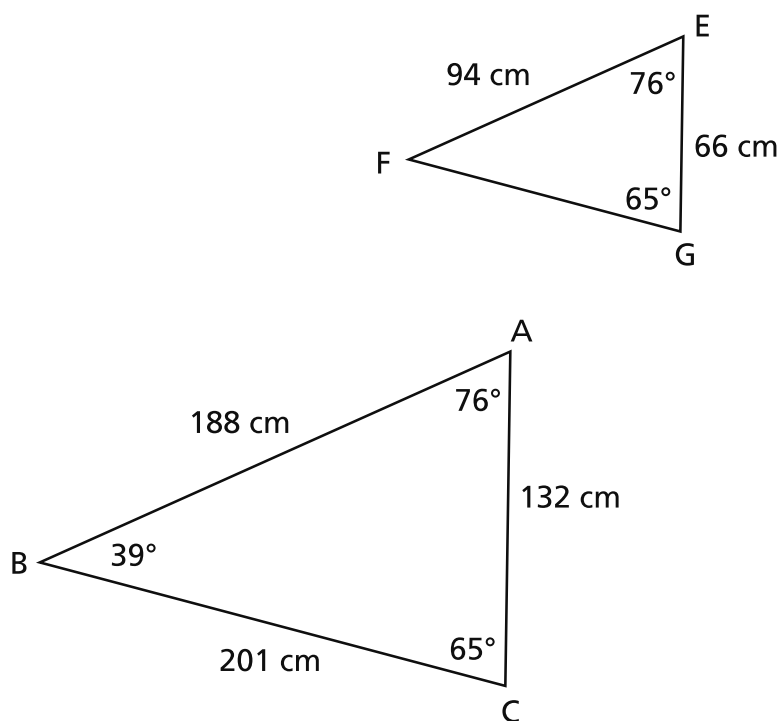
D $y = x^2 - x$

GO ON

15 Which expression is equivalent to $(5^2)(7^{-2})(5^4)$?

- A $\frac{5^6}{7^2}$
- B $\frac{25^6}{7^2}$
- C 17^4
- D 175^4

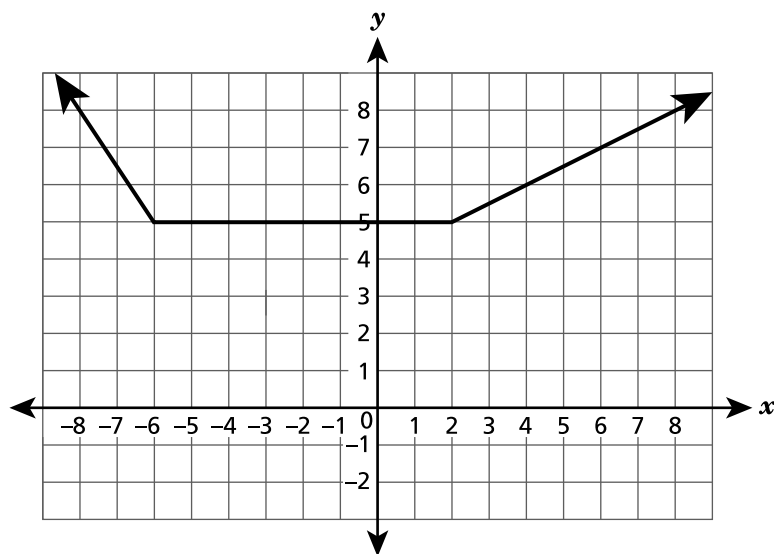
Triangle ABC and triangle EFG are shown below.



Which statement describes the measurements of angle F and side FG for triangle EFG ?

- A Angle F is 39° and side FG is 201 cm.
- B Angle F is 19.5° and side FG is 201 cm.
- C Angle F is 19.5° and side FG is 100.5 cm.
- D Angle F is 39° and side FG is 100.5 cm.

The graph of a function is shown below.



Which statement is true about the function?

- A The function is constant when x is less than -6 .
- B The function is decreasing when x is greater than 2 .
- C The function is increasing when x is greater than 2 .
- D The function is decreasing when x is between the values of -6 and 2 .

Myla sells hot chocolate at a fundraising event. The function $y = 3x + 30$ represents the total amount of money, y , she raises and includes the profit made by selling x cups of hot chocolate and a one time donation at the event. Which statement is true about the situation?

- A The profit per cup is \$3.00 and the number of cups she sells is 30.
- B The number of cups she sells is 3 and the profit per cup is \$30.00.
- C The profit per cup is \$3.00 and the amount of the donation is \$30.00.
- D The amount of the donation is \$3.00 and the profit per cup is \$30.00.

20

Which equation represents the line that passes through the points $(0, 9)$ and $(3, 0)$?

A $y = -3x + 3$

B $y = -3x + 9$

C $y = -\left(\frac{1}{3}\right)x + 3$

D $y = -\left(\frac{1}{3}\right)x + 9$

21

An equation is shown below.

$$3x - 2 = 7(x + 5)$$

What is the value of x ?

A $-\frac{37}{4}$

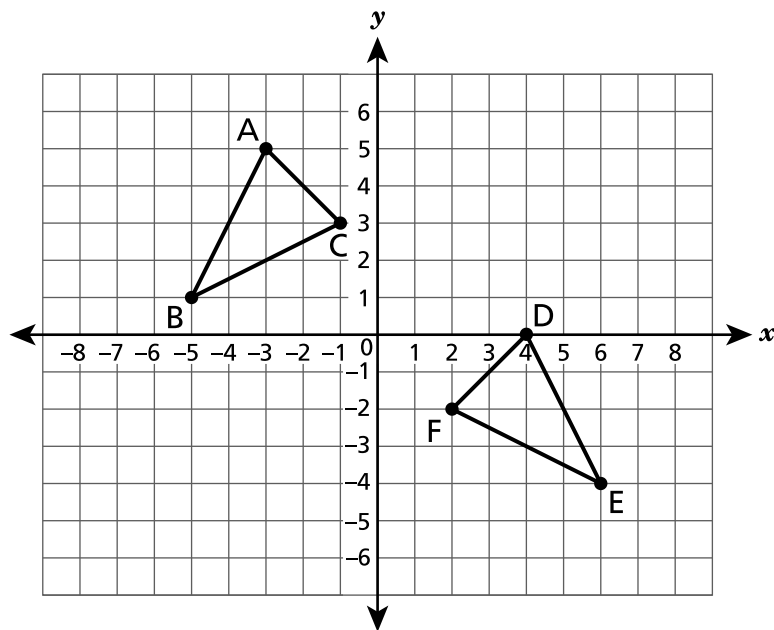
B $-\frac{7}{4}$

C $\frac{3}{10}$

D $\frac{33}{10}$

GO ON

Triangle ABC and triangle DEF shown on the coordinate plane below are congruent.

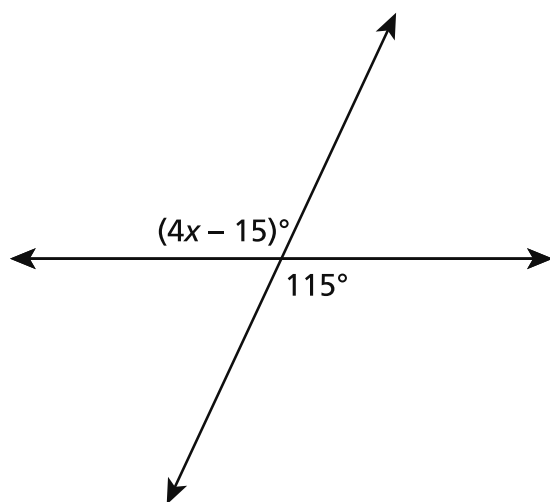


Which sequence of transformations will map triangle ABC onto triangle DEF ?

- A a reflection over the x axis and then a translation 1 unit right and 5 units down
- B a reflection over the x axis and then a translation 5 units right and 1 unit down
- C a reflection over the y axis and then a translation 5 units right and 1 unit down
- D a reflection over the y axis and then a translation 1 unit right and 5 units down

23

Two intersecting lines create the angles shown in the diagram below.



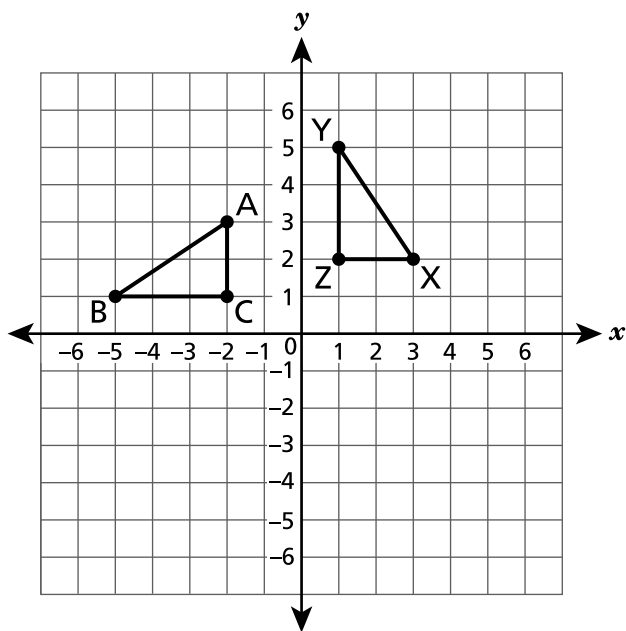
What is the value of x ?

- A 20
- B 26.25
- C 32.5
- D 48.75

GO ON

25

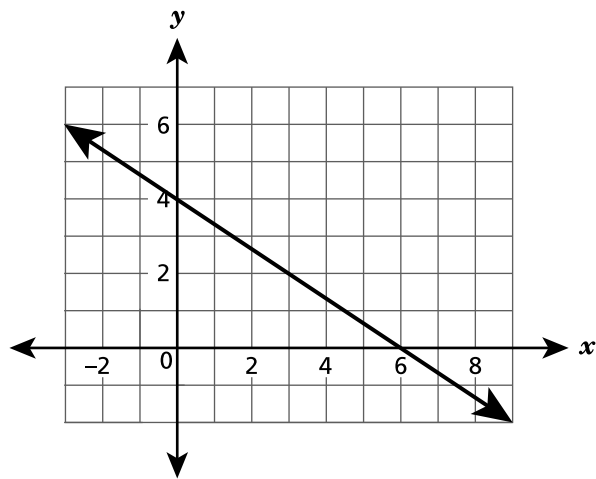
Triangle ABC is rotated clockwise about the origin to create triangle XYZ as shown on the coordinate plane below.



Which statement is correct?

- A $\angle A \cong \angle Y$
- B $\angle B \cong \angle Y$
- C $\angle B \cong \angle Z$
- D $\angle C \cong \angle X$

A linear function is graphed on the coordinate plane shown below.



Which statement is true about the function?

- A The rate of change is $\frac{2}{3}$ and the y intercept is 4.
- B The rate of change is $\frac{2}{3}$ and the y intercept is 6.
- C The rate of change is $-\frac{2}{3}$ and the y intercept is 4.
- D The rate of change is $-\frac{2}{3}$ and the y intercept is 6.

Miguel is comparing the costs of renting a photobooth for an event. The costs of two companies are described below.

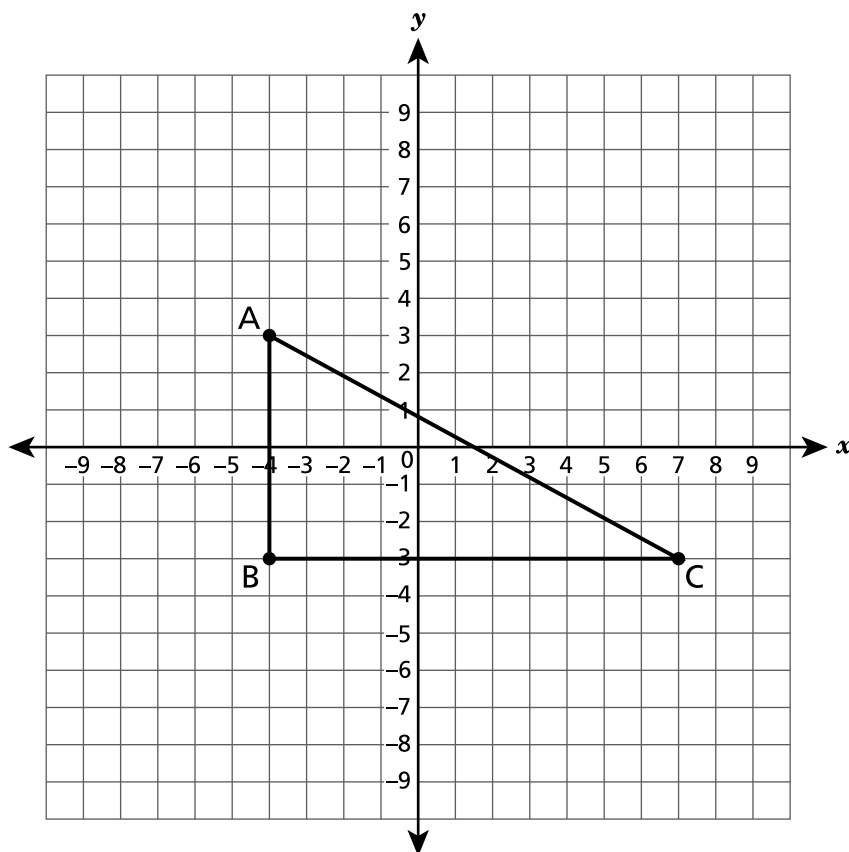
- The cost at Company A includes a \$180.00 set up fee plus an additional \$49.00 per hour.
- The cost at Company B is represented by the equation $c = 99h + 150$, where c is the cost, in dollars, for h hours.

Which statement correctly compares the costs for renting a photobooth?

- A** Company A charges \$30.00 more per hour than Company B.
- B** Company B charges \$50.00 more per hour than Company A.
- C** Company A charges \$50.00 more per hour than Company B.
- D** Company B charges \$30.00 more per hour than Company A.

29

Triangle ABC is graphed on the coordinate plane shown below. Triangle ABC is translated 4 units left and 3 units up to create triangle $A'B'C'$.



What is the length, in units, of side $A'B'$?

- A 2
- B 6
- C 7
- D 9

GO ON

30

A set of ordered pairs is shown below.

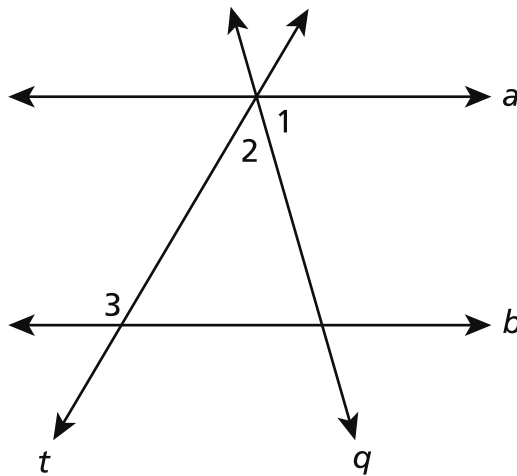
$$\{(1,4), (2,5), (3,4), (k,6)\}$$

Which value of k makes the set of ordered pairs a function?

- A 1
- B 2
- C 3
- D 4

31

In the figure shown below, line a is parallel to line b , and lines q and t are transversals.



In the figure, the measure of angle 1 is 79° and the measure of angle 3 is 128° . What is the measure, in degrees, of angle 2?

- A 49
- B 52
- C 101
- D 128

GO ON

32

An equation is shown below.

$$\frac{1}{2}(14 + 4x) = 5(x - 2) + 2$$

What value of x makes the equation true?

A $2\frac{1}{7}$

B 5

C $6\frac{1}{3}$

D 7

STOP

**Grade 8
Mathematics Test
Session 1
Spring 2026**

Name: _____



New York State Testing Program

Mathematics Test Session 2

Grade 8

Spring 2026

RELEASED QUESTIONS

Session 2



TIPS FOR TAKING THE TEST

Here are some ideas to help you do your best:

- Read each question carefully. Take your time.
- You have a ruler, a protractor, a reference sheet, and a calculator that you can use on the test if they help you answer the question.
- Be sure to show your work when asked.
- Be sure to explain your answer when asked.

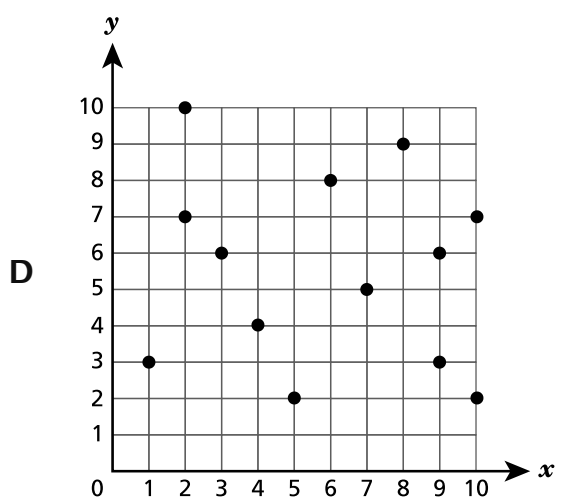
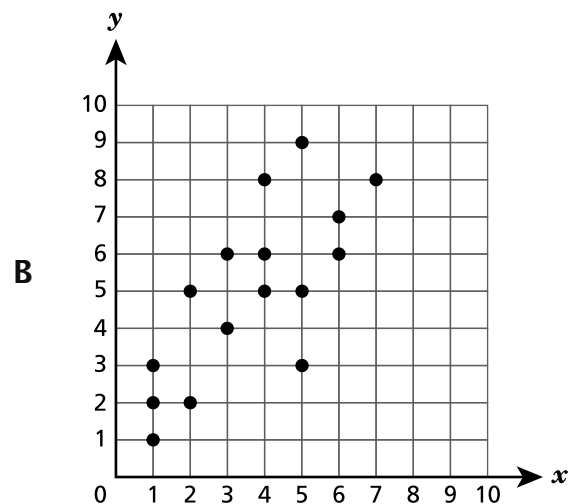
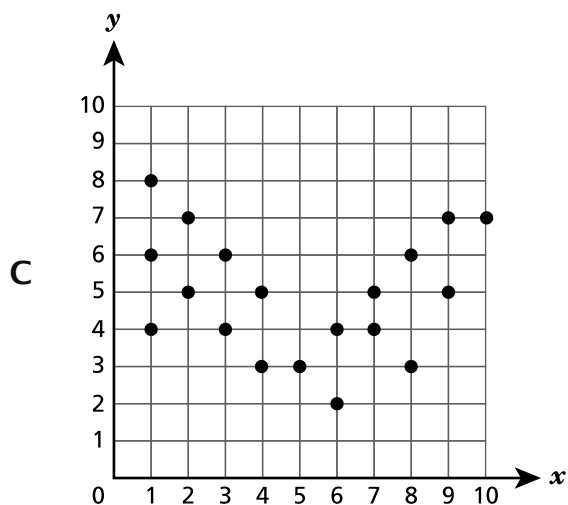
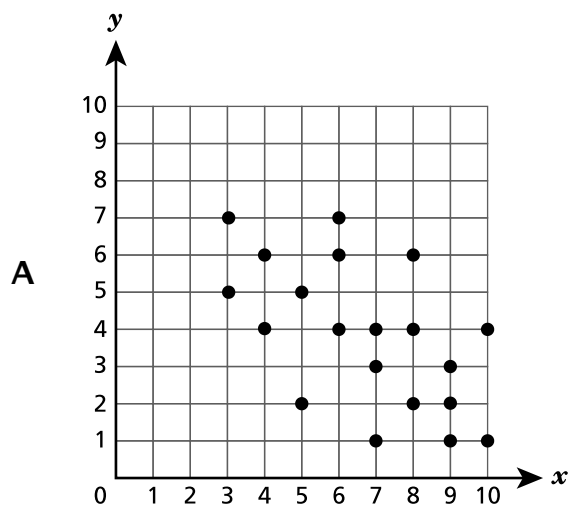
33

Which value of x makes the equation $x^3 = 8$ true?

- A 2
- B 4
- C 11
- D 24

34

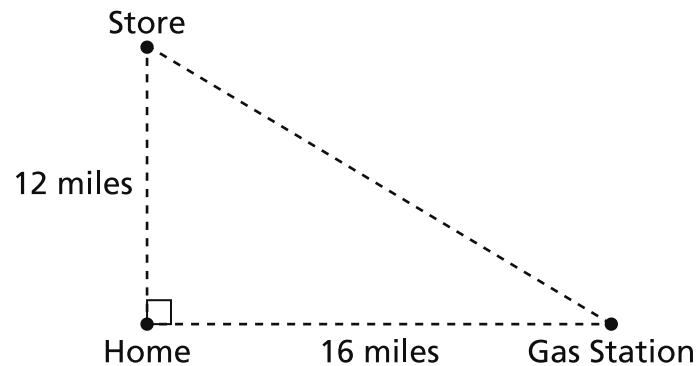
Which scatter plot represents data with a positive linear association?



GO ON

35

The diagram below shows a map of a store, a home, and a gas station. Directly north of the home is a store that is 12 miles away. Directly east of the home is a gas station that is 16 miles away.



What is the distance, in miles, between the store and the gas station?

- A 14
- B 20
- C 28
- D 56

36

An equation is shown below.

$$2(4x + 5) = 5(2x + 1) - 2x$$

Which statement about the equation is true?

- A The equation has no solution.
- B The equation has one solution, $x = \frac{5}{2}$.
- C The equation has one solution, $x = \frac{5}{4}$.
- D The equation has an infinite number of solutions.

GO ON

37

Which table of values represents y as a linear function of x ?

A

x	y
2	4
4	16
6	36
8	64

C

x	y
2	8
4	32
6	72
8	128

B

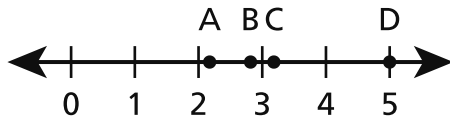
x	y
2	8
4	64
6	216
8	512

D

x	y
2	6
4	10
6	14
8	18

38

Points A, B, C, and D are plotted on the number line shown below.



Which point represents the **best** approximation of the value of $\sqrt{10}$?

A point A

B point B

C point C

D point D

GO ON

40

This question is worth 1 credit.

A cone has a height of 12 inches and a radius of 3.2 inches. What is the volume, in cubic inches, of the cone to the nearest tenth of a cubic inch?

Answer _____ cubic inches

GO ON

This question is worth 1 credit.

Triangle ABC is graphed on a coordinate plane. The coordinates of the vertices are listed below.

- A $(6, -4)$
- B $(-3, 8)$
- C $(-9, 3)$

Triangle ABC is dilated by a scale factor of $\frac{1}{3}$ with the center of dilation at the origin, resulting in Triangle A'B'C'. What are the coordinates of C' after the dilation?

Answer _____

43

This question is worth 2 credits.

A student wrote an incomplete comparison statement for two expressions as shown below.

$$(16^5)^4 \bigcirc 16^8 \cdot 16^{12}$$

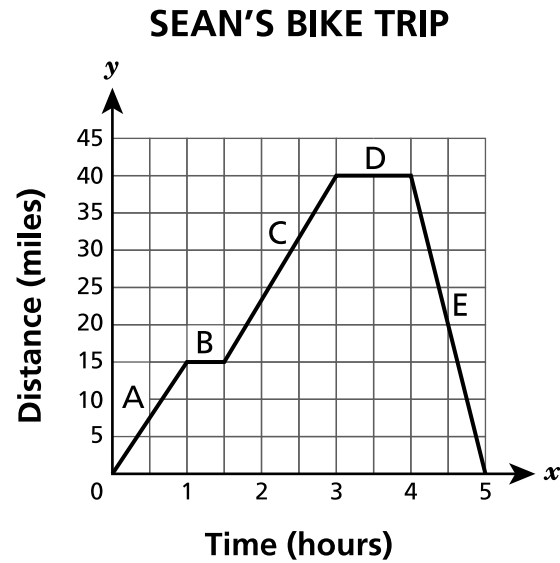
Use the properties of exponents to explain which symbol, $>$, $<$, or $=$, should be placed in the circle to make the comparison statement true.

Explain your answer.

GO ON

This question is worth 2 credits.

Sean is riding his bike along a trail. His distance from the start of the trail is a function of time, as shown in the graph below.



Which segments of the graph show a period of time, if any, where Sean was resting and not riding his bike along the trail?

Explain your answer.

45

This question is worth 2 credits.

Jacqueline builds a box in the shape of a right rectangular prism with dimensions that measure 20.5 inches in length, 13.5 inches in width, and 10 inches in height. What is the volume, in cubic inches, of this box?

Show your work.

Answer _____ cubic inches

GO ON

46

This question is worth 2 credits.

A coffee shop manager tracks the daily high temperature and the number of cups of coffee sold daily in October to predict sales in November. The manager uses this data to model the relationship between the number of cups of coffee sold daily in October, y , and the daily temperature, x , in degrees Fahrenheit with the equation $y = -2.3x + 187$. Based on the model, how many cups of coffee can the manager predict to be sold in one day in November if the high temperature for that day is 60°F ?

Show your work.

Answer _____ cups

GO ON

47

This question is worth 2 credits.

Jack buys a circular rug with a circumference of 5.5π feet. What is the area, to the nearest square foot, of the rug?

Show your work.

Answer _____ square feet

GO ON

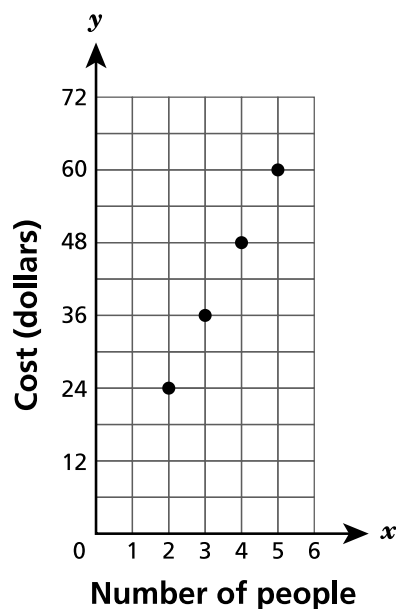
This question is worth 3 credits.

A group of friends is comparing the cost to go water tubing. The cost is proportional to the number of people in the group for two companies, Splash Zone and Tube Time, as shown in the table and the graph below.

SPLASH ZONE

Number of people, x	Cost, c
3	\$33.75
4	\$45.00
5	\$56.25
6	\$67.50

TUBE TIME



What is the difference in the cost, between the two companies, for 8 friends to go tubing? Be sure to include how the values in the table and in the graph were used in your answer.

Explain how you determined your answer.

STOP

THE STATE EDUCATION DEPARTMENT
THE UNIVERSITY OF THE STATE OF NEW YORK / ALBANY, NY 12234
2026 Mathematics Tests Map to the Standards
Grade 8

Question	Type	Key	Points	Standard	Cluster	Subscore	Secondary Standard(s)
Session 1							
1	Multiple Choice	D	1	NGLS.Math.Content.NY-8.G.3	Geometry	Geometry	
2	Multiple Choice	B	1	NGLS.Math.Content.NY-8.EE.5	Expressions and Equations	Expressions and Equations	
3	Multiple Choice	D	1	NGLS.Math.Content.NY-8.EE.1	Expressions and Equations	Expressions and Equations	
4	Multiple Choice	D	1	NGLS.Math.Content.NY-8.SP.3	Statistics and Probability	Statistics and Probability	
5	Multiple Choice	B	1	NGLS.Math.Content.NY-8.NS.1	Number Systems		NGLS.Math.Content.NY-8.EE.2
6	Multiple Choice	C	1	NGLS.Math.Content.NY-8.G.9	Geometry	Geometry	
7	Multiple Choice	D	1	NGLS.Math.Content.NY-8.G.6	Geometry	Geometry	
8	Multiple Choice	A	1	NGLS.Math.Content.NY-8.EE.6	Expressions and Equations	Expressions and Equations	
10	Multiple Choice	D	1	NGLS.Math.Content.NY-8.G.4	Geometry	Geometry	
11	Multiple Choice	D	1	NGLS.Math.Content.NY-8.F.1	Functions	Functions	
12	Multiple Choice	B	1	NGLS.Math.Content.NY-8.EE.7a	Expressions and Equations	Expressions and Equations	
13	Multiple Choice	C	1	NGLS.Math.Content.NY-7.G.2	Geometry	Geometry	
14	Multiple Choice	A	1	NGLS.Math.Content.NY-8.F.3	Functions	Functions	
15	Multiple Choice	A	1	NGLS.Math.Content.NY-8.EE.1	Expressions and Equations	Expressions and Equations	
17	Multiple Choice	D	1	NGLS.Math.Content.NY-8.G.4	Geometry	Geometry	NGLS.Math.Content.NY-8.G.5
18	Multiple Choice	C	1	NGLS.Math.Content.NY-8.F.5	Functions	Functions	
19	Multiple Choice	C	1	NGLS.Math.Content.NY-8.F.4	Functions	Functions	
20	Multiple Choice	B	1	NGLS.Math.Content.NY-8.EE.6	Expressions and Equations	Expressions and Equations	
21	Multiple Choice	A	1	NGLS.Math.Content.NY-8.EE.7b	Expressions and Equations	Expressions and Equations	
22	Multiple Choice	D	1	NGLS.Math.Content.NY-8.G.2	Geometry	Geometry	
23	Multiple Choice	C	1	NGLS.Math.Content.NY-7.G.5	Geometry	Geometry	
25	Multiple Choice	B	1	NGLS.Math.Content.NY-8.G.1b	Geometry	Geometry	
26	Multiple Choice	C	1	NGLS.Math.Content.NY-8.F.4	Functions	Functions	
27	Multiple Choice	B	1	NGLS.Math.Content.NY-8.F.2	Functions	Functions	
29	Multiple Choice	B	1	NGLS.Math.Content.NY-8.G.1a	Geometry	Geometry	
30	Multiple Choice	D	1	NGLS.Math.Content.NY-8.F.1	Functions	Functions	
31	Multiple Choice	A	1	NGLS.Math.Content.NY-8.G.5	Geometry	Geometry	
32	Multiple Choice	B	1	NGLS.Math.Content.NY-8.EE.7b	Expressions and Equations	Expressions and Equations	
Session 2							
33	Multiple Choice	A	1	NGLS.Math.Content.NY-8.EE.2	Expressions and Equations	Expressions and Equations	
34	Multiple Choice	B	1	NGLS.Math.Content.NY-8.SP.1	Statistics and Probability		
35	Multiple Choice	B	1	NGLS.Math.Content.NY-8.G.7	Geometry	Geometry	
36	Multiple Choice	A	1	NGLS.Math.Content.NY-8.EE.7a	Expressions and Equations	Expressions and Equations	
37	Multiple Choice	D	1	NGLS.Math.Content.NY-8.F.3	Functions	Functions	
38	Multiple Choice	C	1	NGLS.Math.Content.NY-8.NS.2	Number Systems		
40	Constructed Response		1	NGLS.Math.Content.NY-8.G.9	Geometry	Geometry	
41	Constructed Response		1	NGLS.Math.Content.NY-8.G.3	Geometry	Geometry	
43	Constructed Response		2	NGLS.Math.Content.NY-8.EE.1	Expressions and Equations	Expressions and Equations	
44	Constructed Response		2	NGLS.Math.Content.NY-8.F.5	Functions	Functions	
45	Constructed Response		2	NGLS.Math.Content.NY-7.G.6	Geometry	Geometry	
46	Constructed Response		2	NGLS.Math.Content.NY-8.SP.3	Statistics and Probability		
47	Constructed Response		2	NGLS.Math.Content.NY-7.G.4	Geometry	Geometry	
48	Constructed Response		3	NGLS.Math.Content.NY-8.EE.5	Expressions and Equations	Expressions and Equations	

*This item map is intended to identify the primary analytic skills necessary to successfully answer each question. However, some questions measure proficiencies described in multiple standards, including a balanced combination of procedural and conceptual understanding.

1-Credit Constructed-Response Rubric

1 Credit	A 1-credit response is a correct answer to the question which indicates a thorough understanding of mathematical concepts and/or procedures.
0 Credits*	A 0-credit response is incorrect, irrelevant, or incoherent.

* Condition Code A is applied whenever a student who is present for a test session leaves an entire constructed-response question in that session completely blank (no response attempted).

2-Credit Constructed-Response Holistic Rubric

2 Credits	A 2-credit response includes the correct solution to the question and demonstrates a thorough understanding of the mathematical concepts and/or procedures in the task. This response • indicates that the student has completed the task correctly, using mathematically sound procedures • contains sufficient work to demonstrate a thorough understanding of the mathematical concepts and/or procedures • may contain inconsequential errors that do not detract from the correct solution and the demonstration of a thorough understanding
1 Credit	A 1-credit response demonstrates only a partial understanding of the mathematical concepts and/or procedures in the task. This response • correctly addresses only some elements of the task • may contain an incorrect solution but applies a mathematically appropriate process • may contain the correct solution but required work is incomplete
0 Credits*	A 0-credit response is incorrect, irrelevant, incoherent, or contains a correct solution obtained using an obviously incorrect procedure. Although some elements may contain correct mathematical procedures, holistically they are not sufficient to demonstrate even a limited understanding of the mathematical concepts embodied in the task.

* Condition Code A is applied whenever a student who is present for a test session leaves an entire constructed-response question in that session completely blank (no response attempted).

3-Credit Constructed-Response Holistic Rubric

3 Credits	A 3-credit response includes the correct solution(s) to the question and demonstrates a thorough understanding of the mathematical concepts and/or procedures in the task. This response • indicates that the student has completed the task correctly, using mathematically sound procedures • contains sufficient work to demonstrate a thorough understanding of the mathematical concepts and/or procedures • may contain inconsequential errors that do not detract from the correct solution(s) and the demonstration of a thorough understanding
2 Credits	A 2-credit response demonstrates a partial understanding of the mathematical concepts and/or procedures in the task. This response • appropriately addresses most but not all aspects of the task using mathematically sound procedures • may contain an incorrect solution but provides sound procedures, reasoning, and/or explanations • may reflect some minor misunderstanding of the underlying mathematical concepts and/or procedures
1 Credit	A 1-credit response demonstrates only a limited understanding of the mathematical concepts and/or procedures in the task. This response • may address some elements of the task correctly but reaches an inadequate solution and/or provides reasoning that is faulty or incomplete • exhibits multiple flaws related to misunderstanding of important aspects of the task, misuse of mathematical procedures, or faulty mathematical reasoning • reflects a lack of essential understanding of the underlying mathematical concepts • may contain the correct solution(s) but required work is limited
0 Credits*	A 0-credit response is incorrect, irrelevant, incoherent, or contains a correct solution obtained using an obviously incorrect procedure. Although some elements may contain correct mathematical procedures, holistically they are not sufficient to demonstrate even a limited understanding of the mathematical concepts embodied in the task.

* Condition Code A is applied whenever a student who is present for a test session leaves an entire constructed-response question in that session completely blank (no response attempted).

1-Credit Constructed-Response Mathematics Scoring Policies

1. The student is **not** required to show work for a 1-credit constructed-response question, therefore, any work shown will **not** be scored. A clearly identified correct response should still receive full credit.
2. If the student clearly identifies a correct answer but fails to write that answer in the answer space, the student should still receive full credit.
3. If the student provides one legible response (and one response only), the rater should score the response, even if it has been crossed out.
4. If the student has written more than one response but has crossed some out, the rater should score only the response that has **not** been crossed out.
5. If the student provides more than one response but does not indicate which response is to be considered the correct response and none have been crossed out, the student shall not receive credit.
6. If the student does not provide the answer in the form as directed in the question, the student will not receive credit.
7. In questions requiring number sentences, the number sentences must be written horizontally.
8. When measuring angles with a protractor, there is a ± 5 degrees deviation allowed of the true measure.
9. Condition Code A is applied whenever a student who is present for a test session leaves an entire constructed-response question in that session completely blank (no response attempted). This is not to be confused with a score of zero wherein the student does respond to part or all of the question, but that work results in a score of zero.

2- and 3-Credit Constructed-Response Mathematics Scoring Policies

1. If a student shows the work in other than a designated “Show your work” or “Explain” area, that work should still be scored.
2. If the question requires students to show their work, and the student shows appropriate work and clearly identifies a correct answer but fails to write that answer in the answer space, the student should still receive full credit.
3. If students are directed to show work or provide an explanation, a correct answer with **no** work shown or **no** explanation provided, receives **no** credit.
4. If students are **not** directed to show work, any work shown will **not** be scored. This applies to questions that do **not** ask for any work and questions that ask for work for one part and do **not** ask for work in another part.
5. If the student provides one legible response (and one response only), the rater should score the response, even if it has been crossed out.
6. If the student has written more than one response but has crossed some out, the rater should score only the response that has **not** been crossed out.
7. If the student provides more than one response, but does not indicate which response is to be considered the correct response and none have been crossed out, the student shall not receive full credit.
8. Trial-and-error responses are **not** subject to Scoring Policy #6 above, since crossing out is part of the trial-and-error process.
9. If a response shows repeated occurrences of the same conceptual error within a question, the conceptual error should **not** be considered more than once in gauging the demonstrated level of understanding.
10. In questions requiring number sentences, the number sentences must be written horizontally.
11. When measuring angles with a protractor, there is a +/- 5 degrees deviation allowed of the true measure.
12. Condition Code A is applied whenever a student who is present for a test session leaves an entire constructed-response question in that session completely blank (no response attempted). This is not to be confused with a score of zero wherein the student does respond to part or all of the question but that work results in a score of zero.

A cone has a height of 12 inches and a radius of 3.2 inches. What is the volume, in cubic inches, of the cone to the nearest tenth of a cubic inch?

Answer _____ cubic inches

EXEMPLARY RESPONSE

40

A cone has a height of 12 inches and a radius of 3.2 inches. What is the volume, in cubic inches, of the cone to the nearest tenth of a cubic inch?

Answer 128.7 cubic inches

GUIDE PAPER 1

40

A cone has a height of 12 inches and a radius of 3.2 inches. What is the volume, in cubic inches, of the cone to the nearest tenth of a cubic inch?

$$V = \frac{1}{3} \pi (3.2^2) \times 12$$

$$V = 128.679...$$

$$V = 128.7$$

Answer

cubic inches

Score Credit 1 (out of 1 credit)

A correct answer is provided.

GUIDE PAPER 2

40

A cone has a height of 12 inches and a radius of 3.2 inches. What is the volume, in cubic inches, of the cone to the nearest tenth of a cubic inch?

Answer cubic inches

Score Credit 1 (out of 1 credit)

A correct answer is provided.

GUIDE PAPER 3

40

A cone has a height of 12 inches and a radius of 3.2 inches. What is the volume, in cubic inches, of the cone to the nearest tenth of a cubic inch?

Answer $\frac{1}{3} \pi (3.2)^2 \times 12 = 128.68$ cubic inches

Score Credit 0 (out of 1 credit)

An incorrect answer is provided.

Triangle ABC is graphed on a coordinate plane. The coordinates of the vertices are listed below.

- A (6, -4)
- B (-3, 8)
- C (-9, 3)

Triangle ABC is dilated by a scale factor of $\frac{1}{3}$ with the center of dilation at the origin, resulting in Triangle A'B'C'. What are the coordinates of C' after the dilation?

Answer _____

EXEMPLARY RESPONSE

41

Triangle ABC is graphed on a coordinate plane. The coordinates of the vertices are listed below.

- A (6, -4)
- B (-3, 8)
- C (-9, 3)

Triangle ABC is dilated by a scale factor of $\frac{1}{3}$ with the center of dilation at the origin, resulting in Triangle A'B'C'. What are the coordinates of C' after the dilation?

Answer (-3, 1)

GUIDE PAPER 1

41

Triangle ABC is graphed on a coordinate plane. The coordinates of the vertices are listed below.

- A (6, -4)
- B (-3, 8)
- C (-9, 3)

Triangle ABC is dilated by a scale factor of $\frac{1}{3}$ with the center of dilation at the origin, resulting in Triangle A'B'C'. What are the coordinates of C' after the dilation?

Answer (-3,1)

Score Credit 1 (out of 1 credit)

A correct answer is provided.

GUIDE PAPER 2

41

Triangle ABC is graphed on a coordinate plane. The coordinates of the vertices are listed below.

- A (6, -4)
- B (-3, 8)
- C (-9, 3)

Triangle ABC is dilated by a scale factor of $\frac{1}{3}$ with the center of dilation at the origin, resulting in Triangle A'B'C'. What are the coordinates of C' after the dilation?

Answer

A prime = (2, -1.3)
B Prime = (-1, 2.6)
C Prime = (-3, 1)

Score Credit 1 (out of 1 credit)

A correct answer is provided.

GUIDE PAPER 3

41

Triangle ABC is graphed on a coordinate plane. The coordinates of the vertices are listed below.

- A (6, -4)
- B (-3, 8)
- C (-9, 3)

Triangle ABC is dilated by a scale factor of $\frac{1}{3}$ with the center of dilation at the origin, resulting in Triangle A'B'C'. What are the coordinates of C' after the dilation?

Answer

Score Credit 0 (out of 1 credit)

An incorrect answer is provided.

A student wrote an incomplete comparison statement for two expressions as shown below.

$$(16^5)^4 \bigcirc 16^8 \cdot 16^{12}$$

Use the properties of exponents to explain which symbol, $>$, $<$, or $=$, should be placed in the circle to make the comparison statement true.

Explain your answer.

EXEMPLARY RESPONSE

43

A student wrote an incomplete comparison statement for two expressions as shown below.

$$(16^5)^4 \bigcirc 16^8 \cdot 16^{12}$$

Use the properties of exponents to explain which symbol, $>$, $<$, or $=$, should be placed in the circle to make the comparison statement true.

Explain your answer.

The symbol $=$ should be placed in the circle.

When a power is raised to a power,
you multiply the exponents and keep the base the same,
so the expression $(16^5)^4$ is equivalent to 16^{20} .
When two expressions with the same base are multiplied,
you keep the base and add the exponents
so the expression $(16^8)(16^{12})$ is also equivalent to 16^{20} .
Since $(16^5)^4$ and $(16^8)(16^{12})$ are each equivalent to 16^{20} ,
they are equal to each other.

OR other valid explanation

A student wrote an incomplete comparison statement for two expressions as shown below.

$$(16^5)^4 \bigcirc 16^8 \cdot 16^{12}$$

Use the properties of exponents to explain which symbol, $>$, $<$, or $=$, should be placed in the circle to make the comparison statement true.

Explain your answer.

The symbol that should be used is $=$ because using the rule power to the power for $(16^5)^4$ the exponents should be multiplied and base stays the same which gets you 16^{20} and using product rule $16^8 \times 16^{12}$ the base stays the same but you add the exponents and get 16^{20} and you get the same 16^{20} on both sides so they are equal so the equal sign.

Score Credit 2 (out of 2 credits)

This response demonstrates a thorough understanding of the mathematical concepts and procedures in the task.

- The correct symbol is chosen and a valid explanation is provided.

This response is complete and correct.

GUIDE PAPER 2

43

A student wrote an incomplete comparison statement for two expressions as shown below.

$$(16^5)^4 \bigcirc 16^8 \cdot 16^{12}$$

Use the properties of exponents to explain which symbol, $>$, $<$, or $=$, should be placed in the circle to make the comparison statement true.

Explain your answer.

Handwritten work showing the simplification of the expression $(16^5)^4$ to 16^{20} using the power of a power property (5×4). The expression $16^8 \cdot 16^{12}$ is simplified to 16^{20} using the product of powers property ($8 + 12$). The final result is $=$.

Score Credit 2 (out of 2 credits)

This response demonstrates a thorough understanding of the mathematical concepts and procedures in the task.

- The correct symbol is chosen, and properties of exponents are correctly used to explain the choice.
- This response is complete and correct.

GUIDE PAPER 3

43

A student wrote an incomplete comparison statement for two expressions as shown below.

$$(16^5)^4 \bigcirc 16^8 \cdot 16^{12}$$

Use the properties of exponents to explain which symbol, $>$, $<$, or $=$, should be placed in the circle to make the comparison statement true.

Explain your answer.

= this sybole would be right because $16^8 \times 16^{12} = 16^{20}$
and $(16^5)^4$ would also become 16^{20}

Score Credit 2 (out of 2 credits)

This response demonstrates a thorough understanding of the mathematical concepts and procedures in the task.

- The correct symbol is chosen and a valid explanation is provided.

This response is sufficient to demonstrate a thorough understanding.

GUIDE PAPER 4

43

A student wrote an incomplete comparison statement for two expressions as shown below.

$$(16^5)^4 \bigcirc 16^8 \cdot 16^{12}$$

Use the properties of exponents to explain which symbol, $>$, $<$, or $=$, should be placed in the circle to make the comparison statement true.

Explain your answer.

The equal symbol would make the comparison statement true because $8 + 12 = 20$ and $5 \times 4 = 20$.

Score Credit 1 (out of 2 credits)

This response demonstrates only a partial understanding of the mathematical concepts and procedures in the task.

- The correct symbol is chosen.
- However, the explanation is incomplete.

This response correctly addresses only some elements of the task.

GUIDE PAPER 5

43

A student wrote an incomplete comparison statement for two expressions as shown below.

$$(16^5)^4 \bigcirc 16^8 \cdot 16^{12}$$

Use the properties of exponents to explain which symbol, $>$, $<$, or $=$, should be placed in the circle to make the comparison statement true.

Explain your answer.

= Because 16 to the 20th power is equal to 16 to the 20th power

Score Credit 1 (out of 2 credits)

This response demonstrates only a partial understanding of the mathematical concepts and procedures in the task.

- The correct symbol is chosen.
- However, the explanation is incomplete.

This response correctly addresses only some elements of the task.

GUIDE PAPER 6

43

A student wrote an incomplete comparison statement for two expressions as shown below.

$$(16^5)^4 \bigcirc 16^8 \cdot 16^{12}$$

Use the properties of exponents to explain which symbol, $>$, $<$, or $=$, should be placed in the circle to make the comparison statement true.

Explain your answer.

i say they are = to because when you multiply exponents with the same base they are added and that would equal 20 and the other one is multipleid by the exponets also =ing to 20

Score Credit 1 (out of 2 credits)

This response demonstrates only a partial understanding of the mathematical concepts and procedures in the task.

- The correct symbol is chosen.
- However, the explanation is incomplete.

This response correctly addresses only some elements of the task.

GUIDE PAPER 7

43

A student wrote an incomplete comparison statement for two expressions as shown below.

$$(16^5)^4 \bigcirc 16^8 \cdot 16^{12}$$

Use the properties of exponents to explain which symbol, $>$, $<$, or $=$, should be placed in the circle to make the comparison statement true.

Explain your answer.

The answer should be $=$ since each side is equal to 1.21

Score Credit 0 (out of 2 credits)

This response is not sufficient to demonstrate even a limited understanding of the mathematical concepts in the task.

- Although the correct symbol is chosen, an incorrect explanation is provided.

Holistically, this response shows no overall understanding of the task.

A student wrote an incomplete comparison statement for two expressions as shown below.

$$(16^5)^4 \bigcirc 16^8 \cdot 16^{12}$$

Use the properties of exponents to explain which symbol, $>$, $<$, or $=$, should be placed in the circle to make the comparison statement true.

Explain your answer.

the symbol $<$ should be used for this answer because if you add $8 + 12 = 20$ then. $16 \times 16 = 256$. but if you add $5 + 4 = 9$ so you will get $16^9 < 256^{20}$

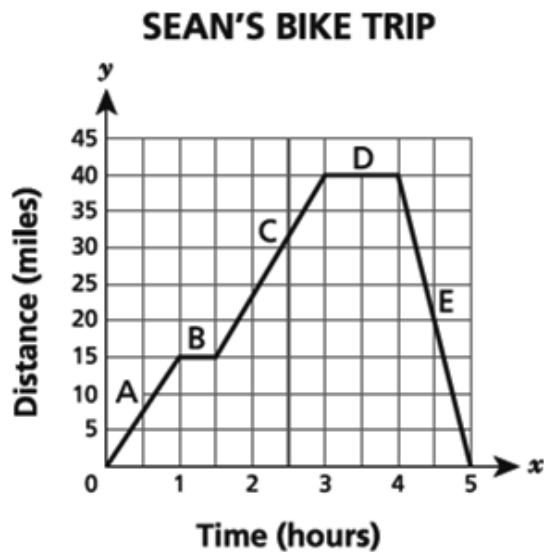
Score Credit 0 (out of 2 credits)

This response is not sufficient to demonstrate even a limited understanding of the mathematical concepts in the task.

- The properties of exponents are applied incorrectly, resulting in an incorrect symbol chosen.

Holistically, this response shows no overall understanding of the task.

Sean is riding his bike along a trail. His distance from the start of the trail is a function of time, as shown in the graph below.



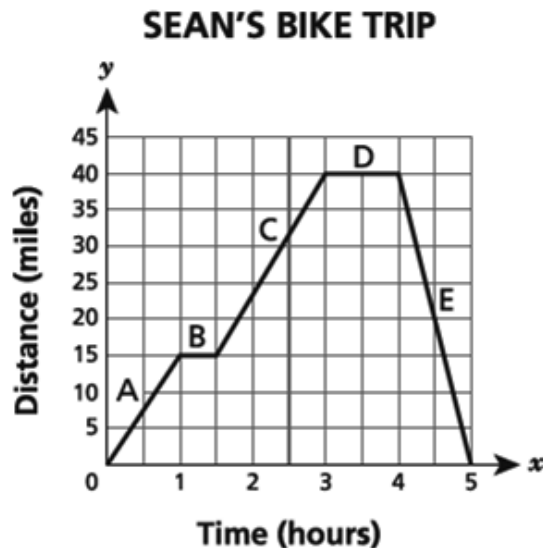
Which segments of the graph show a period of time, if any, where Sean was resting and not riding his bike along the trail?

Explain your answer.

EXEMPLARY RESPONSE

44

Sean is riding his bike along a trail. His distance from the start of the trail is a function of time, as shown in the graph below.



Which segments of the graph show a period of time, if any, where Sean was resting and not riding his bike along the trail?

Explain your answer.

Sean is not bike riding during segments B and D because during those segments the line is horizontal meaning the distance in miles is not changing but time is.

OR other valid explanation

Sean is riding his bike along a trail. His distance from the start of the trail is a function of time, as shown in the graph below.



Which segments of the graph show a period of time, if any, where Sean was resting and not riding his bike along the trail?

Explain your answer.

Segment B and D because the distance is the same for a period of time, and the line is horizontal, meaning that Sean is not moving, but staying in place. For segment b his distance stays at 15 miles for 30 mintues. Also for segment D his distance stays at 40 miles for 1 hour.

Score Credit 2 (out of 2 credits)

This response demonstrates a thorough understanding of the mathematical concepts and procedures in the task.

- The correct segments are identified, and a valid explanation is provided.

This response is complete and correct.

GUIDE PAPER 2

44

Sean is riding his bike along a trail. His distance from the start of the trail is a function of time, as shown in the graph below.



Which segments of the graph show a period of time, if any, where Sean was resting and not riding his bike along the trail?

Explain your answer.

During segments B and D, Sean wasn't riding his bike. I can tell because even though the x variable, time, increases, the y variable, distance stays the same. Therefore, Sean wasn't riding his bike during those time segments.

Score Credit 2 (out of 2 credits)

This response demonstrates a thorough understanding of the mathematical concepts and procedures in the task.

- The correct segments are identified, and a valid explanation is provided.

This response is complete and correct.

GUIDE PAPER 3

44

Sean is riding his bike along a trail. His distance from the start of the trail is a function of time, as shown in the graph below.



Which segments of the graph show a period of time, if any, where Sean was resting and not riding his bike along the trail?

Explain your answer.

B and D because the distance didnt change just the time.

Score Credit 2 (out of 2 credits)

This response demonstrates a thorough understanding of the mathematical concepts and procedures in the task.

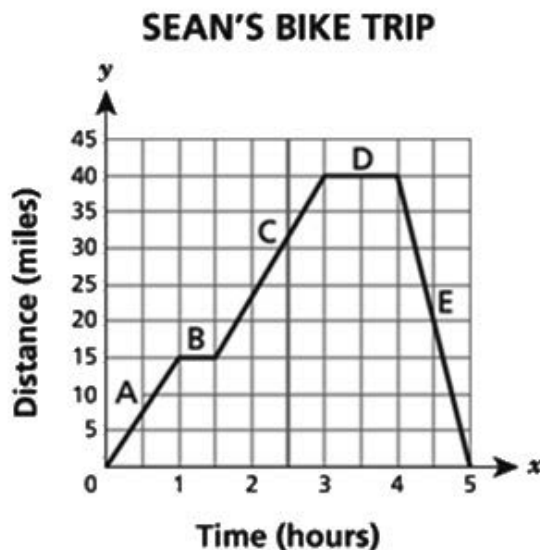
- The correct segments are identified, and a valid explanation is provided.

This response is sufficient to demonstrate a thorough understanding.

GUIDE PAPER 4

44

Sean is riding his bike along a trail. His distance from the start of the trail is a function of time, as shown in the graph below.



Which segments of the graph show a period of time, if any, where Sean was resting and not riding his bike along the trail?

Explain your answer.

B and D because they are the flat points which shows hes not moving.

Score Credit 1 (out of 2 credits)

This response demonstrates only a partial understanding of the mathematical concepts and procedures in the task.

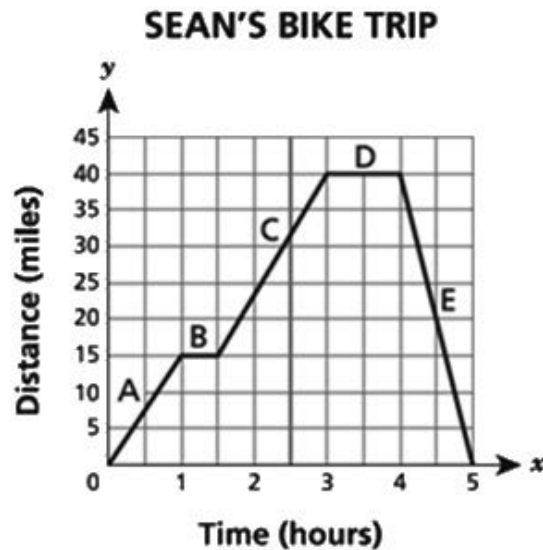
- The correct segments are identified.
- However, the explanation is incomplete.

This response correctly addresses only some elements of the task.

GUIDE PAPER 5

44

Sean is riding his bike along a trail. His distance from the start of the trail is a function of time, as shown in the graph below.



Which segments of the graph show a period of time, if any, where Sean was resting and not riding his bike along the trail?

Explain your answer.

B and D because it's a horizontal line

Score Credit 1 (out of 2 credits)

This response demonstrates only a partial understanding of the mathematical concepts and procedures in the task.

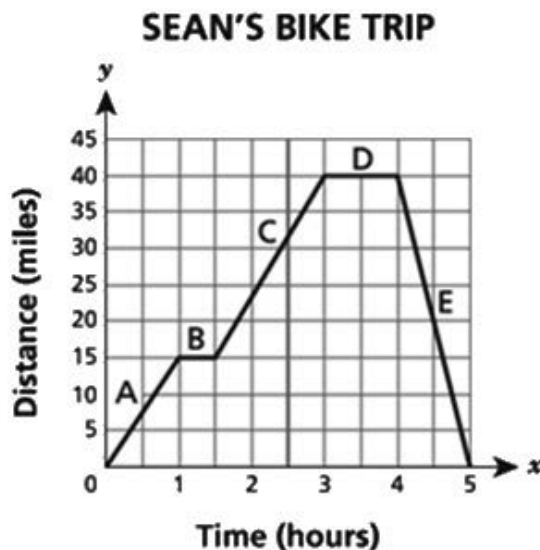
- The correct segments are identified.
- However, the explanation is incomplete.

This response correctly addresses only some elements of the task.

GUIDE PAPER 6

44

Sean is riding his bike along a trail. His distance from the start of the trail is a function of time, as shown in the graph below.



Which segments of the graph show a period of time, if any, where Sean was resting and not riding his bike along the trail?

Explain your answer.

segment D of the graph shows a period of time where sean was resting because the distance stayed the same

Score Credit 1 (out of 2 credits)

This response demonstrates only a partial understanding of the mathematical concepts and procedures in the task.

- Segment D is correctly identified, and a valid explanation is provided.
- However, segment B is not identified.

This response correctly addresses only some elements of the task.

GUIDE PAPER 7

44

Sean is riding his bike along a trail. His distance from the start of the trail is a function of time, as shown in the graph below.



Which segments of the graph show a period of time, if any, where Sean was resting and not riding his bike along the trail?

Explain your answer.

Sean was resting in B and D becuase you can see where he made the stops.

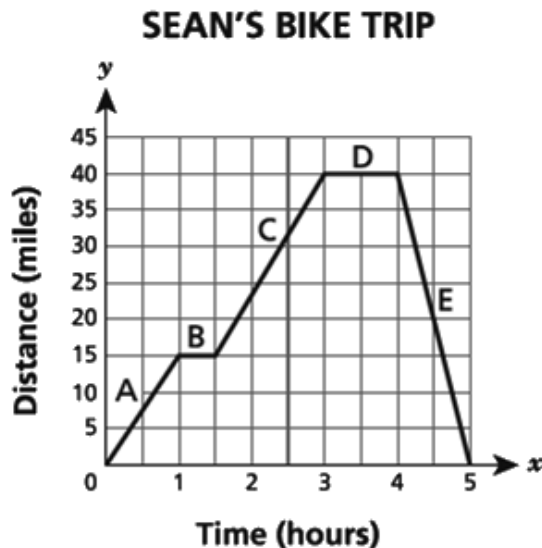
Score Credit 0 (out of 2 credits)

This response is not sufficient to demonstrate even a limited understanding of the mathematical concepts in the task.

- Although the correct segments are identified, the explanation is insufficient.

Holistically, this response shows no overall understanding of the task.

Sean is riding his bike along a trail. His distance from the start of the trail is a function of time, as shown in the graph below.



Which segments of the graph show a period of time, if any, where Sean was resting and not riding his bike along the trail?

Explain your answer.

from hour 1, to 1 1/2. Another is from hour 3 to hour 4.

Score Credit 0 (out of 2 credits)

This response is not sufficient to demonstrate even a limited understanding of the mathematical concepts in the task.

- Although correct intervals of time are stated for the segments, no explanation is provided.

Per Scoring Policy #3 for 2- and 3-credit responses, this response receives no credit.

Jacqueline builds a box in the shape of a right rectangular prism with dimensions that measure 20.5 inches in length, 13.5 inches in width, and 10 inches in height. What is the volume, in cubic inches, of this box?

Show your work.

Answer _____ cubic inches

EXEMPLARY RESPONSE

45

Jacqueline builds a box in the shape of a right rectangular prism with dimensions that measure 20.5 inches in length, 13.5 inches in width, and 10 inches in height. What is the volume, in cubic inches, of this box?

Show your work.

$$V = Bh$$

$$V = (l \times w) \times h$$

$$V = (20.5 \times 13.5) \times 10$$

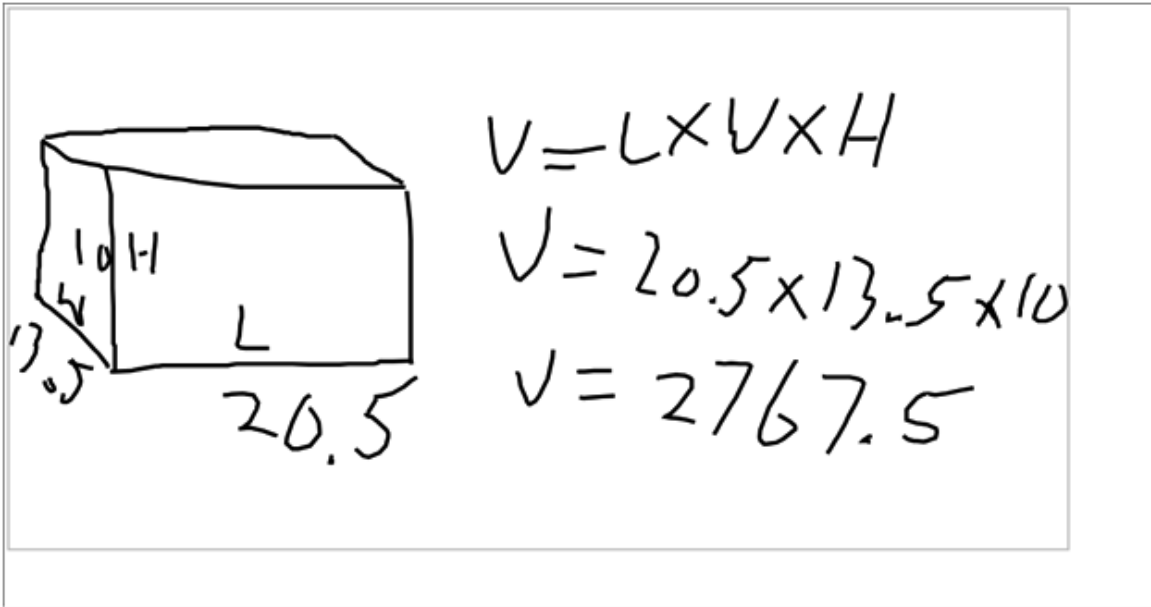
$$V = 2,767.5 \text{ cubic inches}$$

Or other valid process

Answer 2,767.5 cubic inches

Jacqueline builds a box in the shape of a right rectangular prism with dimensions that measure 20.5 inches in length, 13.5 inches in width, and 10 inches in height. What is the volume, in cubic inches, of this box?

Show your work.



Answer cubic inches

Score Credit 2 (out of 2 credits)

This response demonstrates a thorough understanding of the mathematical concepts and procedures in the task.

- The correct volume is determined using a mathematically sound procedure.

This response is complete and correct.

GUIDE PAPER 2

45

Jacqueline builds a box in the shape of a right rectangular prism with dimensions that measure 20.5 inches in length, 13.5 inches in width, and 10 inches in height. What is the volume, in cubic inches, of this box?

Show your work.

$$\begin{aligned} &20.5 \times 13.5 \times 10 \\ &13.5 \times 10 = 135 \\ &135 \times 20.5 = 2767.5 \end{aligned}$$

Answer cubic inches

Score Credit 2 (out of 2 credits)

This response demonstrates a thorough understanding of the mathematical concepts and procedures in the task.

- The correct volume is determined using a mathematically sound procedure.

This response is complete and correct.

GUIDE PAPER 3

45

Jacqueline builds a box in the shape of a right rectangular prism with dimensions that measure 20.5 inches in length, 13.5 inches in width, and 10 inches in height. What is the volume, in cubic inches, of this box?

Show your work.

$$20.5 \times 13.5 \times 10$$

Answer cubic inches

Score Credit 2 (out of 2 credits)

This response demonstrates a thorough understanding of the mathematical concepts and procedures in the task.

- The correct volume is determined using a mathematically sound procedure.

This response is complete and correct.

GUIDE PAPER 4

45

Jacqueline builds a box in the shape of a right rectangular prism with dimensions that measure 20.5 inches in length, 13.5 inches in width, and 10 inches in height. What is the volume, in cubic inches, of this box?

Show your work.

$$\begin{array}{r} 20.5 \times 13.5 \times 10 \\ = \\ 27675 \text{ in}^3 \end{array}$$

Answer cubic inches

Score Credit 1 (out of 2 credits)

This response demonstrates only a partial understanding of the mathematical concepts and procedures in the task.

- A mathematically sound procedure is used to determine the volume.
- However, a calculation error results in an incorrect solution.

This response contains an incorrect solution but applies a mathematically appropriate process.

GUIDE PAPER 5

45

Jacqueline builds a box in the shape of a right rectangular prism with dimensions that measure 20.5 inches in length, 13.5 inches in width, and 10 inches in height. What is the volume, in cubic inches, of this box?

Show your work.

$l \times w \times h$

2767.5

Answer 2767.5 cubic inches

Score Credit 1 (out of 2 credits)

This response demonstrates only a partial understanding of the mathematical concepts and procedures in the task.

- The correct formula for volume is displayed, with the correct volume shown.
- However, it is unclear how this solution was determined.

This response contains the correct solution, but the required work is incomplete.

GUIDE PAPER 6

45

Jacqueline builds a box in the shape of a right rectangular prism with dimensions that measure 20.5 inches in length, 13.5 inches in width, and 10 inches in height. What is the volume, in cubic inches, of this box?

Show your work.

$$20.5 \times 13.5 = 276.75$$

Answer

276.75

cubic inches

Score Credit 1 (out of 2 credits)

This response demonstrates only a partial understanding of the mathematical concepts and procedures in the task.

- The base area of the box is correctly determined.
- However, no further work is shown to find the volume, and the base area of the box is inappropriately provided as the solution.

This response correctly addresses only some elements of the task.

GUIDE PAPER 7

45

Jacqueline builds a box in the shape of a right rectangular prism with dimensions that measure 20.5 inches in length, 13.5 inches in width, and 10 inches in height. What is the volume, in cubic inches, of this box?

Show your work.

$$20.5 + 13.5 + 10 = 44$$

Answer 44 cubic inches cubic inches

Score Credit 0 (out of 2 credits)

This response is not sufficient to demonstrate even a limited understanding of the mathematical concepts in the task.

- An incorrect procedure is used to determine an incorrect solution.

Holistically, this response shows no overall understanding of the task.

45

Jacqueline builds a box in the shape of a right rectangular prism with dimensions that measure 20.5 inches in length, 13.5 inches in width, and 10 inches in height. What is the volume, in cubic inches, of this box?

Show your work.

we basically just have to multiply all the numbers together

Answer 27675 cubic inches

Score Credit 0 (out of 2 credits)

This response is not sufficient to demonstrate even a limited understanding of the mathematical concepts in the task.

- An incorrect solution is provided, and the description of the procedure is insufficient.

Holistically, this response shows no overall understanding of the task.

A coffee shop manager tracks the daily high temperature and the number of cups of coffee sold daily in October to predict sales in November. The manager uses this data to model the relationship between the number of cups of coffee sold daily in October, y , and the daily temperature, x , in degrees Fahrenheit with the equation $y = -2.3x + 187$. Based on the model, how many cups of coffee can the manager predict to be sold in one day in November if the high temperature for that day is 60°F ?

Show your work.

Answer _____ cups

EXEMPLARY RESPONSE

46

A coffee shop manager tracks the daily high temperature and the number of cups of coffee sold daily in October to predict sales in November. The manager uses this data to model the relationship between the number of cups of coffee sold daily in October, y , and the daily temperature, x , in degrees Fahrenheit with the equation $y = -2.3x + 187$. Based on the model, how many cups of coffee can the manager predict to be sold in one day in November if the high temperature for that day is 60°F ?

Show your work.

$$\begin{aligned}y &= -2.3(60) + 187 \\&= -138 + 187 \\&= 49\end{aligned}$$

OR other valid process

Answer 49 cups

A coffee shop manager tracks the daily high temperature and the number of cups of coffee sold daily in October to predict sales in November. The manager uses this data to model the relationship between the number of cups of coffee sold daily in October, y , and the daily temperature, x , in degrees Fahrenheit with the equation $y = -2.3x + 187$. Based on the model, how many cups of coffee can the manager predict to be sold in one day in November if the high temperature for that day is 60°F ?

Show your work.

Y= Cups of coffee sold daily

X=Temperature

$$-2.3(60) = -138$$

$$-138 + 187 = 49$$

The manager can predict to sell 49 cups of coffee.

Answer 49 cups cups

Score Credit 2 (out of 2 credits)

This response demonstrates a thorough understanding of the mathematical concepts and procedures in the task.

- The predicted number of cups of coffee sold on a 60-degree day is correctly determined using mathematically sound procedures.

The response is complete and correct.

GUIDE PAPER 2

46

A coffee shop manager tracks the daily high temperature and the number of cups of coffee sold daily in October to predict sales in November. The manager uses this data to model the relationship between the number of cups of coffee sold daily in October, y , and the daily temperature, x , in degrees Fahrenheit with the equation $y = -2.3x + 187$. Based on the model, how many cups of coffee can the manager predict to be sold in one day in November if the high temperature for that day is 60°F ?

Show your work.

49 cups of coffee can be sold on a day that is 60 degrees because $-2.3 \times 60 + 187 = 49$

Answer 49 cups

Score Credit 2 (out of 2 credits)

This response demonstrates a thorough understanding of the mathematical concepts and procedures in the task.

- The predicted number of cups of coffee sold on a 60-degree day is correctly determined using mathematically sound procedures.

The response is complete and correct.

GUIDE PAPER 3

46

A coffee shop manager tracks the daily high temperature and the number of cups of coffee sold daily in October to predict sales in November. The manager uses this data to model the relationship between the number of cups of coffee sold daily in October, y , and the daily temperature, x , in degrees Fahrenheit with the equation $y = -2.3x + 187$. Based on the model, how many cups of coffee can the manager predict to be sold in one day in November if the high temperature for that day is 60°F ?

Show your work.

$$-2.3(60)+187=49$$

Answer cups

Score Credit 2 (out of 2 credits)

This response demonstrates a thorough understanding of the mathematical concepts and procedures in the task.

- The predicted number of cups of coffee sold on a 60-degree day is correctly determined using mathematically sound procedures.

The response is complete and correct.

GUIDE PAPER 4

46

A coffee shop manager tracks the daily high temperature and the number of cups of coffee sold daily in October to predict sales in November. The manager uses this data to model the relationship between the number of cups of coffee sold daily in October, y , and the daily temperature, x , in degrees Fahrenheit with the equation $y = -2.3x + 187$. Based on the model, how many cups of coffee can the manager predict to be sold in one day in November if the high temperature for that day is 60°F ?

Show your work.

$$\begin{aligned} & -2.3(60)+187 \\ & -2.3 \times 60=138 \\ & 138+187=325 \text { which is the cups sold } \end{aligned}$$

Answer cups

Score Credit 1 (out of 2 credits)

This response demonstrates only a partial understanding of the mathematical concepts and procedures in the task.

- A mathematically sound procedure is used to determine the predicted number of cups of coffee to be sold.
- However, a calculation error ($-2.3 \times 60 = 138$) occurs, resulting in an incorrect solution.

This response contains an incorrect solution but applies a mathematically appropriate process.

GUIDE PAPER 5

46

A coffee shop manager tracks the daily high temperature and the number of cups of coffee sold daily in October to predict sales in November. The manager uses this data to model the relationship between the number of cups of coffee sold daily in October, y , and the daily temperature, x , in degrees Fahrenheit with the equation $y = -2.3x + 187$. Based on the model, how many cups of coffee can the manager predict to be sold in one day in November if the high temperature for that day is 60°F ?

Show your work.

$$\begin{aligned} y &= -2.3x + 187 \\ y &= -2.3(60) + 187 \\ y &= -138 + 187 \\ y &= 49 \text{ cups} \end{aligned}$$
$$\begin{array}{r} 184 \\ - 49 \\ \hline 135 \text{ cups sold} \end{array}$$

Answer

135

cups

Score Credit 1 (out of 2 credits)

This response demonstrates only a partial understanding of the mathematical concepts and procedures in the task.

- A mathematically sound procedure is used to determine the predicted number of cups of coffee to be sold.
- However, the correct predicted number of cups is inappropriately subtracted from 184, resulting in an incorrect solution.

This response contains an incorrect solution but applies a mathematically appropriate process.

GUIDE PAPER 6

46

A coffee shop manager tracks the daily high temperature and the number of cups of coffee sold daily in October to predict sales in November. The manager uses this data to model the relationship between the number of cups of coffee sold daily in October, y , and the daily temperature, x , in degrees Fahrenheit with the equation $y = -2.3x + 187$. Based on the model, how many cups of coffee can the manager predict to be sold in one day in November if the high temperature for that day is 60°F ?

Show your work.

$$\begin{array}{l} y = -2.3x + 187 \\ \quad \downarrow \\ -2.3 \cdot 60 = -150 \\ 187 - 150 = 37 \end{array}$$

Answer cups

Score Credit 1 (out of 2 credits)

This response demonstrates only a partial understanding of the mathematical concepts and procedures in the task.

- A mathematically sound procedure is used to determine the predicted number of cups of coffee to be sold.
- However, a calculation error ($-2.3 \times 60 = -150$) occurs, resulting in an incorrect solution.

This response contains an incorrect solution but applies a mathematically appropriate process.

GUIDE PAPER 7

46

A coffee shop manager tracks the daily high temperature and the number of cups of coffee sold daily in October to predict sales in November. The manager uses this data to model the relationship between the number of cups of coffee sold daily in October, y , and the daily temperature, x , in degrees Fahrenheit with the equation $y = -2.3x + 187$. Based on the model, how many cups of coffee can the manager predict to be sold in one day in November if the high temperature for that day is 60°F ?

Show your work.

$$2.3x + 187 + 60 = 249.3$$

Answer cups

Score Credit 0 (out of 2 credits)

This response is not sufficient to demonstrate even a limited understanding of the mathematical concepts in the task.

- An incorrect procedure is used to find an incorrect solution.

Holistically, this response shows no overall understanding of the task.

46

A coffee shop manager tracks the daily high temperature and the number of cups of coffee sold daily in October to predict sales in November. The manager uses this data to model the relationship between the number of cups of coffee sold daily in October, y , and the daily temperature, x , in degrees Fahrenheit with the equation $y = -2.3x + 187$. Based on the model, how many cups of coffee can the manager predict to be sold in one day in November if the high temperature for that day is 60°F ?

Show your work.

$$187 \times 0.23x = 43.01$$

Answer cups

Score Credit 0 (out of 2 credits)

This response is not sufficient to demonstrate even a limited understanding of the mathematical concepts in the task.

- An incorrect procedure is used, and an incorrect solution is provided.

Holistically, this response shows no overall understanding of the task.

Jack buys a circular rug with a circumference of 5.5π feet. What is the area, to the nearest square foot, of the rug?

Show your work.

Answer _____ square feet

EXEMPLARY RESPONSE

47

Jack buys a circular rug with a circumference of 5.5π feet. What is the area, to the nearest square foot, of the rug?

Show your work.

$$C = \pi d$$

$$5.5\pi = \pi d$$

$$5.5 = d$$

$$r = 5.5/2 = 2.75$$

$$A = \pi r^2$$

$$A = \pi(2.75)^2$$

$$A = \pi(7.5625)$$

$$A = 23.75829... \approx 24$$

OR other valid process

Answer 24 square feet

47

Jack buys a circular rug with a circumference of 5.5π feet. What is the area, to the nearest square foot, of the rug?

Show your work.

$$5.5\pi \div \pi = 5.5$$

5.5=diameter

$$5.5 \div 2 = 2.75(\text{radius})$$

$$A = \pi \times r^2$$

$$A = \pi \times 2.75^2 = 23.758.... \text{ rounds to } = 24 \text{ square feet}$$

about 24 square feet

Answer

square feet

Score Credit 2 (out of 2 credits)

This response demonstrates a thorough understanding of the mathematical concepts and procedures in the task.

- The area of the rug, to the nearest square foot, is correctly determined using a mathematically sound procedure.

This response is complete and correct.

GUIDE PAPER 2

47

Jack buys a circular rug with a circumference of 5.5π feet. What is the area, to the nearest square foot, of the rug?

Show your work.

$$C = 2\pi r$$

$$5.5\pi = 2\pi r$$

$$/2 \quad /2$$

$$2.75\pi = \pi r$$

$$/\pi \quad /\pi$$

$$2.75 = r$$

$$A = \pi r^2$$

$$A = \pi 2.75^2$$

$$a = 23.76$$

Answer square feet

Score Credit 2 (out of 2 credits)

This response demonstrates a thorough understanding of the mathematical concepts and procedures in the task.

- The area of the rug, to the nearest square foot, is correctly determined using a mathematically sound procedure.

This response contains sufficient work to demonstrate a thorough understanding.

GUIDE PAPER 3

47

Jack buys a circular rug with a circumference of 5.5π feet. What is the area, to the nearest square foot, of the rug?

Show your work.

$$c = d \times \pi \quad A = \pi \times r^2$$

$$d = 5.5$$

$$r = 2.75$$

$$A = 2.75^2 \pi \quad A = 7.5625 \times \pi = 23.75829444$$

Answer square feet

Score Credit 2 (out of 2 credits)

This response demonstrates a thorough understanding of the mathematical concepts and procedures in the task.

- The area of the rug, to the nearest square foot, is correctly determined using a mathematically sound procedure.

This response contains sufficient work to demonstrate a thorough understanding.

GUIDE PAPER 4

47

Jack buys a circular rug with a circumference of 5.5π feet. What is the area, to the nearest square foot, of the rug?

Show your work.

area of a circle = πr^2
circumference = πd
radius = $1/2$ of diameter
 $5.5\pi \div \pi = 5.5$
 $5.5 = \text{diameter}$
 $1/2$ of $5.5 = 2.75 = \text{radius}$
 $\pi \times 2.75 \times 2.75 = 7.5625... \pi$
 $= 8\pi$

Answer square feet

Score Credit 1 (out of 2 credits)

This response demonstrates only a partial understanding of the mathematical concepts and procedures in the task.

- A mathematically correct procedure is used to determine the area of the rug.
- However, inappropriate rounding occurs, and the answer is left in terms of π , resulting in an incorrect solution.

This response correctly addresses only some elements of the task.

GUIDE PAPER 5

47

Jack buys a circular rug with a circumference of 5.5π feet. What is the area, to the nearest square foot, of the rug?

Show your work.

$$A = \pi r^2$$

$$5.5 \div 2 = 2.75$$

$$A = \pi 2.75^2$$

$$A = 23.75829444$$

$$A = 23.8$$

Answer

23.8

square feet

Score Credit 1 (out of 2 credits)

This response demonstrates only a partial understanding of the mathematical concepts and procedures in the task.

- A mathematically correct procedure is used to determine the area of the rug.
- However, the answer is not rounded to the nearest square foot, resulting in an incorrect solution.

This response correctly addresses only some elements of the task.

GUIDE PAPER 6

47

Jack buys a circular rug with a circumference of 5.5π feet. What is the area, to the nearest square foot, of the rug?

Show your work.

$$5.5/2=2.25$$

$$2.25 \text{ squared}=5.0625$$

$$5.0625 \times \pi =15.90431...$$

Answer square feet

Score Credit 1 (out of 2 credits)

This response demonstrates only a partial understanding of the mathematical concepts and procedures in the task.

- A mathematically correct procedure is used to determine the area of the rug.
- However, a calculation error in determining the radius occurs, resulting in an incorrect solution.

This response contains an incorrect solution but applies a mathematically appropriate process.

GUIDE PAPER 7

47

Jack buys a circular rug with a circumference of 5.5π feet. What is the area, to the nearest square foot, of the rug?

Show your work.

$$5.5 \times 3.14 = 17.27$$

17 sqaure foot

Answer 17 square feet

Score Credit 0 (out of 2 credits)

This response is not sufficient to demonstrate even a limited understanding of the mathematical concepts in the task.

- An incorrect procedure is used to determine an incorrect solution.

Holistically, this response shows no overall understanding of the task.

47

Jack buys a circular rug with a circumference of 5.5π feet. What is the area, to the nearest square foot, of the rug?

Show your work.

5.5 divided by 2 is 2.75

Answer 6.4 square feet

Score Credit 0 (out of 2 credits)

This response is not sufficient to demonstrate even a limited understanding of the mathematical concepts in the task.

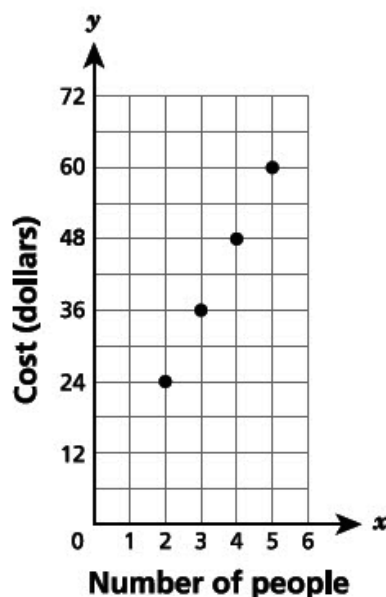
- Although the radius is correctly determined, no further calculations are performed, and an incorrect solution is provided.

Holistically, this response is insufficient to show any understanding.

A group of friends is comparing the cost to go water tubing. The cost is proportional to the number of people in the group for two companies, Splash Zone and Tube Time, as shown in the table and the graph below.

SPLASH ZONE

Number of people, x	Cost, c
3	\$33.75
4	\$45.00
5	\$56.25
6	\$67.50

TUBE TIME

What is the difference in the cost, between the two companies, for 8 friends to go tubing? Be sure to include how the values in the table and in the graph were used in your answer.

Explain how you determined your answer.

EXEMPLARY RESPONSE

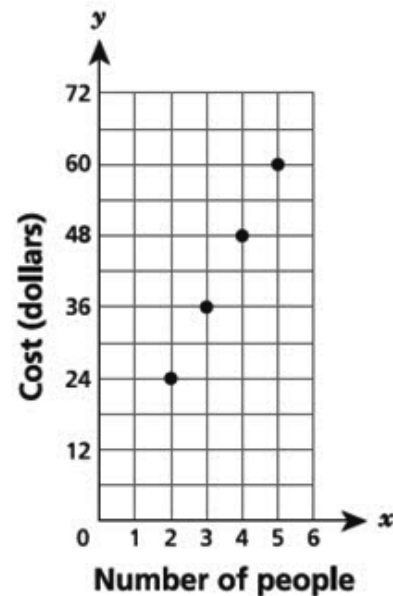
48

A group of friends is comparing the cost to go water tubing. The cost is proportional to the number of people in the group for two companies, Splash Zone and Tube Time, as shown in the table and the graph below.

SPLASH ZONE

Number of people, x	Cost, c
3	\$33.75
4	\$45.00
5	\$56.25
6	\$67.50

TUBE TIME



What is the difference in the cost, between the two companies, for 8 friends to go tubing? Be sure to include how the values in the table and in the graph were used in your answer.

Explain how you determined your answer.

For Splash Zone, the unit rate is the constant of proportionality (cost per person), so $33.75 \div 3$ is \$11.25 per person.

For Tube Time, the slope of the graph is $12/1$ which means that for every 1 person, the price increases by 12.

So the unit rate is \$12.00.

For 8 friends at Splash Zone, the cost would be $8 \times 11.25 = \$90.00$.

And for 8 friends at Tube Time the cost would be $8 \times 12.00 = \$96.00$.

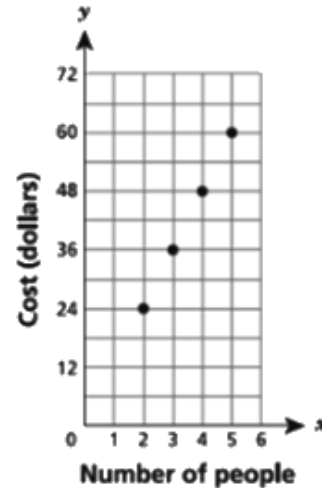
So the difference in the cost between the 2 companies for 8 friends is $96 - 90$ which is equal to \$6.00.

OR other valid explanation

A group of friends is comparing the cost to go water tubing. The cost is proportional to the number of people in the group for two companies, Splash Zone and Tube Time, as shown in the table and the graph below.

SPLASH ZONE

Number of people, x	Cost, c
3	\$33.75
4	\$45.00
5	\$56.25
6	\$67.50

TUBE TIME

What is the difference in the cost, between the two companies, for 8 friends to go tubing? Be sure to include how the values in the table and in the graph were used in your answer.

Explain how you determined your answer.

The difference, in the cost, between the two companies, for 8 friends to go tubing is \$6. To find the RoC of Splash Zone, you can use the points (3, 33.75) and (4, 45) and then use $\frac{Y_2 - Y_1}{X_2 - X_1}$, which would get you $\frac{45 - 33.75}{4 - 3}$, which is equal to 11.25, so when you plug it into the equation $Y = Mx + B$, you would get $Y = 11.25x$. For Tube Time, you can use the points (2, 24) and (3, 36), and then use $\frac{Y_2 - Y_1}{X_2 - X_1}$, which would get you $\frac{36 - 24}{3 - 2}$, which is equal to 12, so when you plug it into the equation, you would get $Y = 12x$. Now you can plug 8 for X in both equations to $Y = 90$ for Splash Zone, and $Y = 96$ for Tube Time and when you subtract 90 from 96, you would get 6.

Score Credit 3 (out of 3 credits)

This response demonstrates a thorough understanding of the mathematical concepts and procedures in the task.

- The correct difference in cost for 8 people to go tubing is determined, and a valid explanation is provided.

This response is complete and correct.

GUIDE PAPER 2

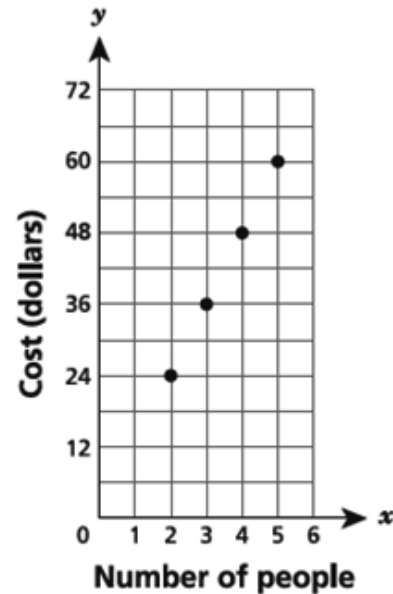
48

A group of friends is comparing the cost to go water tubing. The cost is proportional to the number of people in the group for two companies, Splash Zone and Tube Time, as shown in the table and the graph below.

SPLASH ZONE

Number of people, x	Cost, c
3	\$33.75
4	\$45.00
5	\$56.25
6	\$67.50

TUBE TIME



What is the difference in the cost, between the two companies, for 8 friends to go tubing? Be sure to include how the values in the table and in the graph were used in your answer.

Explain how you determined your answer.

splash zone $33.75 \div 3 = 11.25$ tube time $24 \div 2 = 12$
 $11.25 \times 8 = 90$ $12 \times 8 = 96$
theres a 6 dollar difference

Score Credit 3 (out of 3 credits)

This response demonstrates a thorough understanding of the mathematical concepts and procedures in the task.

- The correct difference in cost for 8 people to go tubing is determined, and a valid explanation is provided.

This response is complete and correct.

GUIDE PAPER 3

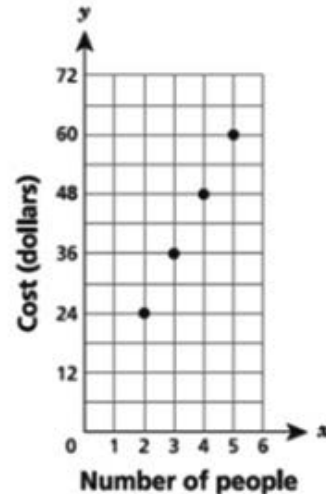
48

A group of friends is comparing the cost to go water tubing. The cost is proportional to the number of people in the group for two companies, Splash Zone and Tube Time, as shown in the table and the graph below.

SPLASH ZONE

Number of people, x	Cost, c
3	\$33.75
4	\$45.00
5	\$56.25
6	\$67.50

TUBE TIME



What is the difference in the cost, between the two companies, for 8 friends to go tubing? Be sure to include how the values in the table and in the graph were used in your answer.

Explain how you determined your answer.

$$67.5 - 11.25 = 56.25 - 11.25 = 45 - 11.25 = 33.75$$

$$24 \div 2 = 12 \quad 36 \div 3 = 12 \quad 48 \div 4 = 12 \quad 60 \div 5 = 12$$

$$11.25 \times 8 = 90$$

$$12 \times 8 = 96$$

$$96 - 90 = 6$$

Score Credit 3 (out of 3 credits)

This response demonstrates a thorough understanding of the mathematical concepts and procedures in the task.

- The correct difference in cost for 8 people to go tubing is determined, and a valid explanation is provided.
- The run-on equation does not detract from the demonstration of a thorough understanding.

This response is sufficient to demonstrate a thorough understanding.

GUIDE PAPER 4

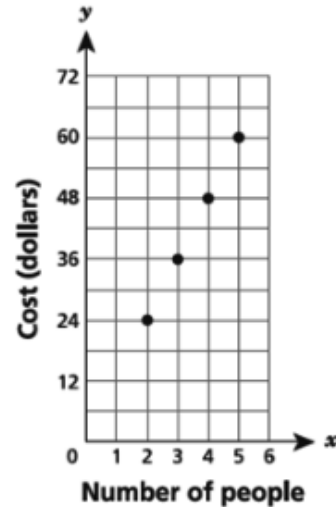
48

A group of friends is comparing the cost to go water tubing. The cost is proportional to the number of people in the group for two companies, Splash Zone and Tube Time, as shown in the table and the graph below.

SPLASH ZONE

Number of people, x	Cost, c
3	\$33.75
4	\$45.00
5	\$56.25
6	\$67.50

TUBE TIME



What is the difference in the cost, between the two companies, for 8 friends to go tubing? Be sure to include how the values in the table and in the graph were used in your answer.

Explain how you determined your answer.

Splash Zone
 $y = 11.25x$
 $y = 11.25(8)$
 $y = 90$

Tube Time
 $y = 12x$
 $y = 12(8)$
 $y = 96$

The difference
 $96 - 90 = 6$

The difference is by 6 dollars and Splash Zone is cheaper than Tube Time.

Score Credit 2 (out of 3 credits)

This response demonstrates a partial understanding of the mathematical concepts and procedures in the task.

- The correct difference in cost for 8 people to go tubing is determined.
- However, it is unclear how the unit rates were determined.

This response appropriately addresses most, but not all, aspects of the task.

GUIDE PAPER 5

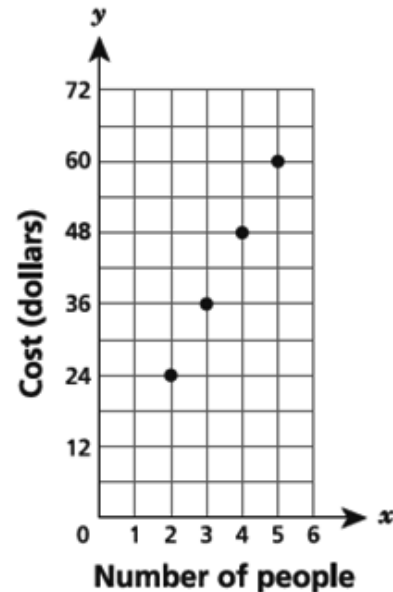
48

A group of friends is comparing the cost to go water tubing. The cost is proportional to the number of people in the group for two companies, Splash Zone and Tube Time, as shown in the table and the graph below.

SPLASH ZONE

Number of people, x	Cost, c
3	\$33.75
4	\$45.00
5	\$56.25
6	\$67.50

TUBE TIME



What is the difference in the cost, between the two companies, for 8 friends to go tubing? Be sure to include how the values in the table and in the graph were used in your answer.

Explain how you determined your answer.

Tube time = 12dollars per person
Splash zone= 11.25 dollars per person
 $(12 \times 8) - (11.25 \times 8) = 6$ dollar difference.

Score Credit 2 (out of 3 credits)

This response demonstrates a partial understanding of the mathematical concepts and procedures in the task.

- The correct difference in cost for 8 people to go tubing is determined.
- However, it is unclear how the unit rates were determined.

This response appropriately addresses most, but not all, aspects of the task.

GUIDE PAPER 6

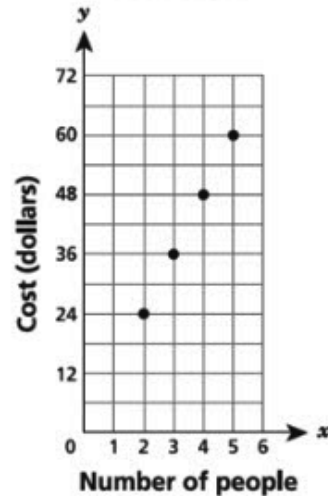
48

A group of friends is comparing the cost to go water tubing. The cost is proportional to the number of people in the group for two companies, Splash Zone and Tube Time, as shown in the table and the graph below.

SPLASH ZONE

Number of people, x	Cost, c
3	\$33.75
4	\$45.00
5	\$56.25
6	\$67.50

TUBE TIME



What is the difference in the cost, between the two companies, for 8 friends to go tubing? Be sure to include how the values in the table and in the graph were used in your answer.

Explain how you determined your answer.

- At Splash Zone the cost per person is \$11.25, which means for 8 people the total cost would be \$90.00
- At Tube Time the cost per person is \$12.00, which means for 8 people the total cost would be \$96.00
- So for 8 people Spalsh Zone costs \$6.00 less than Tube Time

Score Credit 2 (out of 3 credits)

This response demonstrates a partial understanding of the mathematical concepts and procedures in the task.

- The correct difference in cost for 8 people to go tubing is determined.
- However, it is unclear how the unit rates were determined.

This response appropriately addresses most, but not all, aspects of the task.

GUIDE PAPER 7

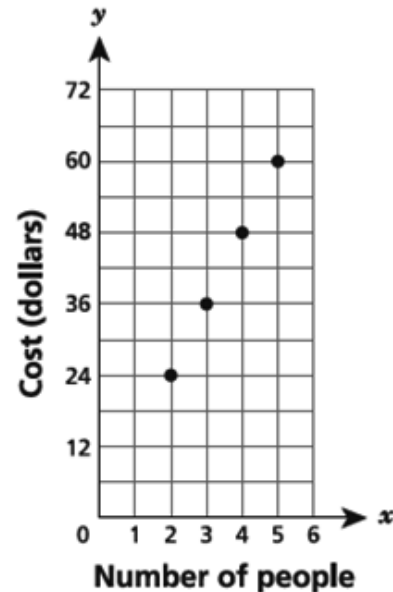
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A group of friends is comparing the cost to go water tubing. The cost is proportional to the number of people in the group for two companies, Splash Zone and Tube Time, as shown in the table and the graph below.

SPLASH ZONE

Number of people, x	Cost, c
3	\$33.75
4	\$45.00
5	\$56.25
6	\$67.50

TUBE TIME



What is the difference in the cost, between the two companies, for 8 friends to go tubing? Be sure to include how the values in the table and in the graph were used in your answer.

Explain how you determined your answer.

Splash zone is \$90 for 8 people and Tube time is \$96 for 8 people which means that there is a \$6 difference between the both.

Score Credit 1 (out of 3 credits)

This response demonstrates only a limited understanding of the mathematical concepts and procedures in the task.

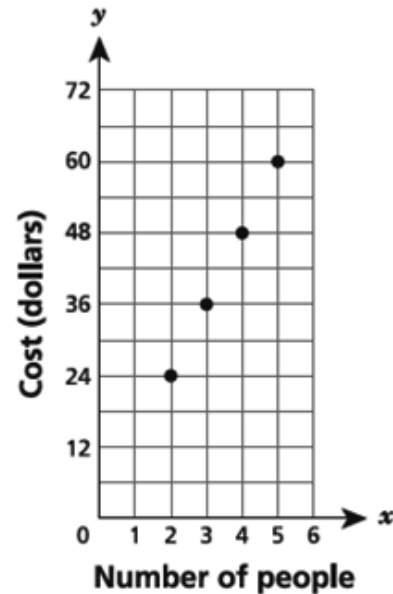
- The correct difference in cost for 8 people to go tubing is determined.
- However, the explanation is incomplete; it is unclear how the total costs were determined.

This response contains the correct solution(s), but the required work is limited.

A group of friends is comparing the cost to go water tubing. The cost is proportional to the number of people in the group for two companies, Splash Zone and Tube Time, as shown in the table and the graph below.

SPLASH ZONE

Number of people, x	Cost, c
3	\$33.75
4	\$45.00
5	\$56.25
6	\$67.50

TUBE TIME

What is the difference in the cost, between the two companies, for 8 friends to go tubing? Be sure to include how the values in the table and in the graph were used in your answer.

Explain how you determined your answer.

The difference in coast is \$6 for 8 people to go. For tube time its \$12 a person and splash zone is \$11.25 a person.

Score Credit 1 (out of 3 credits)

This response demonstrates only a limited understanding of the mathematical concepts and procedures in the task.

- The difference in cost for 8 people to go tubing is correctly provided; however, it is unclear how this was determined.
- The total costs were not addressed, and it is unclear how the unit rates were determined.

This response addresses some elements of the task correctly but provides reasoning that is incomplete.

GUIDE PAPER 9

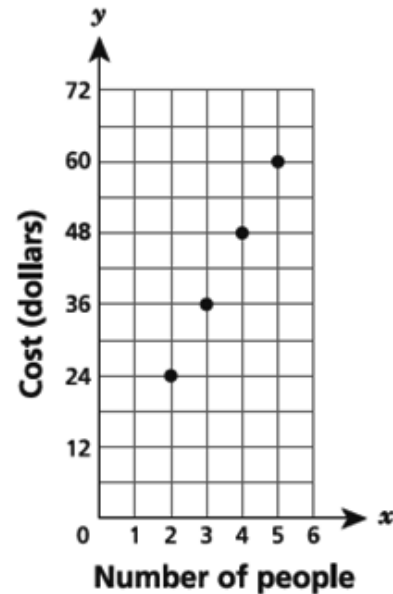
48

A group of friends is comparing the cost to go water tubing. The cost is proportional to the number of people in the group for two companies, Splash Zone and Tube Time, as shown in the table and the graph below.

SPLASH ZONE

Number of people, x	Cost, c
3	\$33.75
4	\$45.00
5	\$56.25
6	\$67.50

TUBE TIME



What is the difference in the cost, between the two companies, for 8 friends to go tubing? Be sure to include how the values in the table and in the graph were used in your answer.

Explain how you determined your answer.

Splash zone = 11.25 per person

Tube time = 12 per person

The difference is \$.75 between the 2 companies.

Score Credit 1 (out of 3 credits)

This response demonstrates only a limited understanding of the mathematical concepts and procedures in the task.

- The unit rate for each company is correctly determined, with the difference between the rates provided.
- However, the explanation is incomplete, it is unclear how the unit rates were determined, and the total costs and difference between the total costs are not addressed.

This response addresses some elements of the task correctly but provides reasoning that is incomplete.

GUIDE PAPER 10

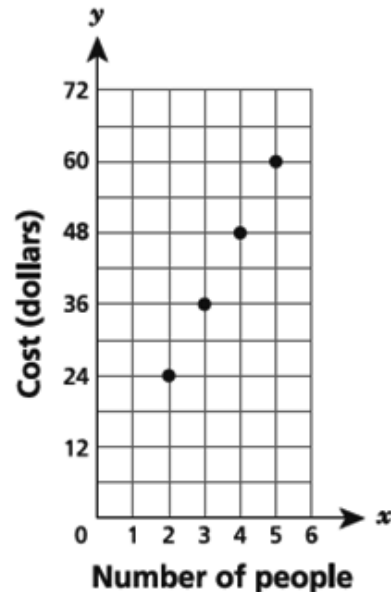
48

A group of friends is comparing the cost to go water tubing. The cost is proportional to the number of people in the group for two companies, Splash Zone and Tube Time, as shown in the table and the graph below.

SPLASH ZONE

Number of people, x	Cost, c
3	\$33.75
4	\$45.00
5	\$56.25
6	\$67.50

TUBE TIME



What is the difference in the cost, between the two companies, for 8 friends to go tubing? Be sure to include how the values in the table and in the graph were used in your answer.

Explain how you determined your answer.

It would be about 87 Or 90 something dollors and for the graph it would be 80 the differents is 10 or less.

Score Credit 0 (out of 3 credits)

This response is not sufficient to demonstrate even a limited understanding of the mathematical concepts and procedures in the task.

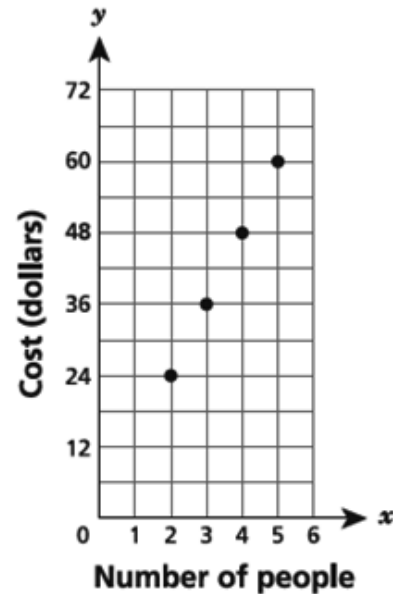
- An incorrect solution is provided with incorrect reasoning.

This response is incorrect, and, holistically, is insufficient to show any understanding.

A group of friends is comparing the cost to go water tubing. The cost is proportional to the number of people in the group for two companies, Splash Zone and Tube Time, as shown in the table and the graph below.

SPLASH ZONE

Number of people, x	Cost, c
3	\$33.75
4	\$45.00
5	\$56.25
6	\$67.50

TUBE TIME

What is the difference in the cost, between the two companies, for 8 friends to go tubing? Be sure to include how the values in the table and in the graph were used in your answer.

Explain how you determined your answer.

The difference between 8 friends tubing between the two companies is \$6

Score Credit 0 (out of 3 credits)

This response is not sufficient to demonstrate even a limited understanding of the mathematical concepts and procedures in the task.

- The correct difference in cost for 8 people to go tubing is provided, with no explanation.

Per Scoring Policy #3 for 2- and 3-credit responses, this response receives no credit.